Search and Unemployment

Raffaele Saggio

UBC

November 1, 2026

- What is unemployment?
- Concept arose in the 1930s from the US statistics agency in the aftermath of the Great depression.

Measurament

Ironically, the questions on unemployment were dropped from the 1920 Census, which was fielded just as a major recession took hold of the U.S. economy. The severity of the downturn led Secretary of Commerce Herbert Hoover to establish a special commission on unemployment, whose findings were reported in an NBER book (Committee of the President's Conference on Unemployment, 1923). Reflecting the poor state of knowledge on labor market statistics, estimates of the number of unemployed at the height of the recession ranged from 3.5 to 5.5 million (van Kleeck, 1923). As an outgrowth of the commission, the American Statistical Association established a Committee on Governmental Labor Statistics (CGLS), which sponsored a series of studies on the collection of labor market statistics (e.g., Hurlin and Berridge, 1926), and played an important role in the design of the questions for the 1930 Census.[4]

Getting/Losing a Job is a BIG DEAL

Table 1 Self-reported important life events in past year: UK data.

	All	Men	Women
Family	38	33	42
Employment	22	24	20
Nothing	20	22	18
Leisure	19	19	19
Education	13	11	15
Health	12	10	13
Consumption	9	9	8
Housing	8	7	9
Other	7	6	7
Financial	4	4	4

Source: British household panel study.

Why there is unemployment?

- Assume an intertemporal labor supply function of the form (Lucas and Rapping, 1969).

$$logh_{it} = \alpha_i + x'_{it}\beta + \eta logw_{it} + \epsilon_{it} \tag{1}$$

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- This leads to one potential notion of unemployment

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- When wages are lower than expected, retract labor supply and that goes into leisure.
- (Some) Macro guys believe that $\eta = 4 \rightarrow$ small reduction in wages leads to take a vacation in Greece for a month.
- No space for the concept of "unemployed" defined as someone actively searching for a job (and is unable to find it!).

Unemployment ≈ Leisure in Neo-classical Labor Supply model

Leisure Luxuries and the Labor Supply of Young Men*

Mark Aguiar Mark Bils Kerwin Kofi Charles Erik Hurst

February 22, 2018

Abstract

Since 2004, time-use data show that younger men, ages 21 to 30, shifted their leisure sharply to video gaming and other recreational computer activities. Over the same period, these younger men exhibited a larger decline in work hours than older men or women. We propose a framework to answer whether improved leisure technology affected younger men's labor supply. The starting point is a leisure demand system that parallels that often estimated for consumer expenditures. We show that total leisure demand is especially sensitive to innovations in leisure luxuries, that is, activities that display a disproportionate response to changes in total leisure time. Using cross-region variation, we estimate that gaming/recreational computer use is distinctly a leisure luxury for younger men. We calculate that innovations to gaming/recreational computing reduced younger men's labor supply by 1.5 to 3.1 percent since 2004. That would explain 23 to 46 percent of their decline in market hours over the period.

Measurament ahead of Theory

e. Measurement Ahead of Theory?

Another possible reason for the late adoption of the search-based definition of unemployment is that social scientists (and economists in particular) had not yet developed a theory of job search. In the prevailing supply-demand framework of the time, a person was either working or not: there was no distinct role for search. In fact, judging by the content of the leading academic journals, unemployment was not an important topic of research interest among economists until well after World War II. In the 30 year period from 1920 to 1949, the *American Economic Review* published only 16 articles with "unemployment", "unemployed" or "labor force" in the title in the regular pages of the journal (21 more appeared in the *Papers and Proceedings*). Over the same period the *Journal of Political Economy* published 14 articles and the *Quarterly Journal of Economics* 19 articles.[16] By comparison 40 articles appeared in the *Journal of the American Statistical Association*, including many papers written by WPA and Census Bureau staff.

Search

- Search models are an alternative (and possibly more reasonable) way to explain why there is unemployment.
- Plan:
  - Brief discussion of search models and concept of the reservation wage.
  - The role of Unemployment insurance.
  - The long-term consequences of becoming unemployed.

Section 1

Search and General Facts About Unemployment

McCall (1970)

- Individuals enter the labor market as unemployed and receive some benefit while unemployed given by b. They live forever and are risk neutral with discount factor $\beta = \dfrac{1}{1+r} < 1$.
- Each period they search and receive a job offer that can be accepted/rejected. Search has (monetary) cost ν.
- Once job is accepted it lasts forever. There is no on the job search (and utility is linear in money).
- Q: what is the p.d.v of working?

Value Functions

- Let U denote the optimal utility of a worker who is however currently unemployed.
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$$\underbrace{U}_{\text{Value of being unemployed}} = b - \nu + \beta \int_0^{\infty} \max\{U, \frac{w}{1-\beta}\} f(w) d(w) \quad (4)$$

- Now define w^* has the level of wage such that $\frac{w^*}{1-\beta} = U$
- We call w^* the reservation wage: lowest level of wage that will make the worker exit unemployment state.

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- We call w^* the reservation wage: lowest level of wage that will make the worker exit unemployment state.
- Useful expression that implicity defines the reservation wage:

$$w^* = b - \nu + \frac{1}{r} \mathrm{E}[w - w^* | w \geq w^*] Pr(w \geq w^*) \tag{5}$$

Unemployment

- All workers here have the same reservation wage.
- Therefore, the probability to leave the unemployment state is given by

$$Pr(w \geq w^*) = 1 - F(w^*) \tag{6}$$

- Now the policy question: what happens when we increase unemployment benefits?

$$\frac{\partial w^*}{\partial b} = \frac{1}{1 + \frac{1}{r}(1 - F(w^*))} > 0 \tag{7}$$

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- Hence, generous unemployment benefits should make searchers picky, raise accepted wages, and lengthen unemployment durations.

Comments

- Constant reservation wage.
- Hard to get data on offer distribution.
- Partial equilibrium model (why?).

Krueger and Mueller (2016)

- How important are reservation wages, really?
- Survey roughly 6,000 unemployed workers in New Jersey over the period October 2009 through March 2010.
- Data available online but low (10%) response rate
- Study how reservation wages should vary with unemployment duration using non-stationary search model ala Mortensen (1977).
- Find that actual behavior is quite different.

Model

- The value function of an unemployed worker with t months of benefits left (there are T months of benefits that the worker can have) is:

$$U(t) = u(b(t)) + \beta \max_R \left\{ U(t-1) + \alpha \int_R (W(x, m=0) - U(t-1)) \, dF(x) \right\}$$

where R is the reservation wage, and $W(w, m=0)$ is the value of starting a job paying wage w.

- To qualify for UI benefits, workers must stay with their employer for $\bar{m} = 6$ months. The value in month $m < \bar{m}$ of a job paying wage w is:

$$\begin{aligned} W(w,m) &= u(w) + \beta \left[(1-\delta) W(w, m+1) + \delta U(0) \right] \\ &\quad + \beta \left[\alpha_e (1-\delta) \int_w (W(x, m+1) - W(w, m+1)) \, dF(x) \right] \end{aligned}$$

where α_e is the probability of finding another job while employed.

- In months, $m \geq \bar{m}$, the worker has qualified for benefits and the problem becomes stationary with:

$$\begin{aligned} W(w,\bar{m}) &= u(w) + \beta \left[(1-\delta) W(w, \bar{m}) + \delta U(T) \right] \\ &\quad + \beta \alpha_e (1-\delta) \int_w (W(x, \bar{m}) - W(w, \bar{m})) \, dF(x) \end{aligned}$$

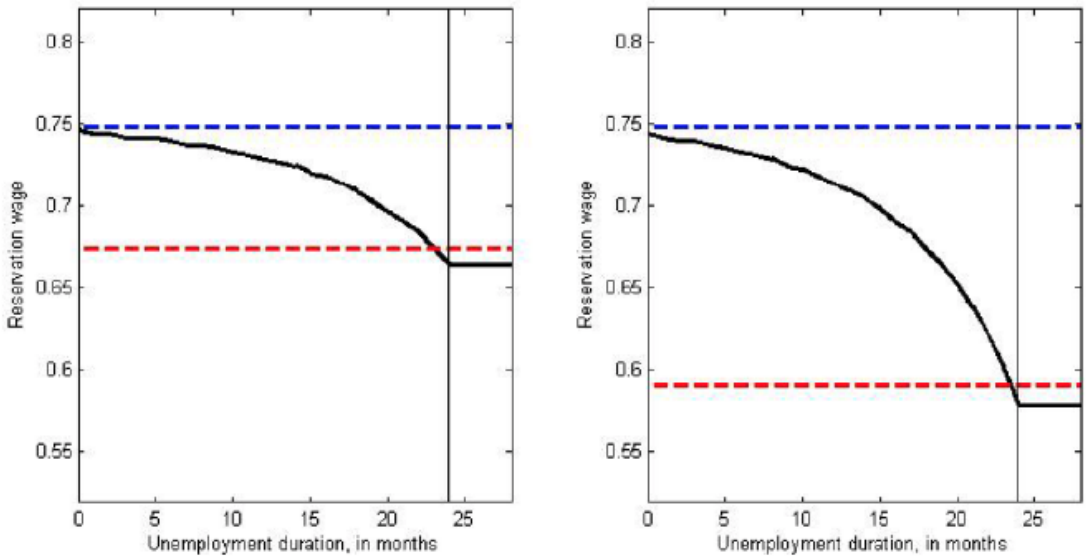
Model

- The reservation wage function $R(t)$ solves the following equation:

$$W(R(t), m = 0) = U(t-1)$$

- No closed form, solve on computer

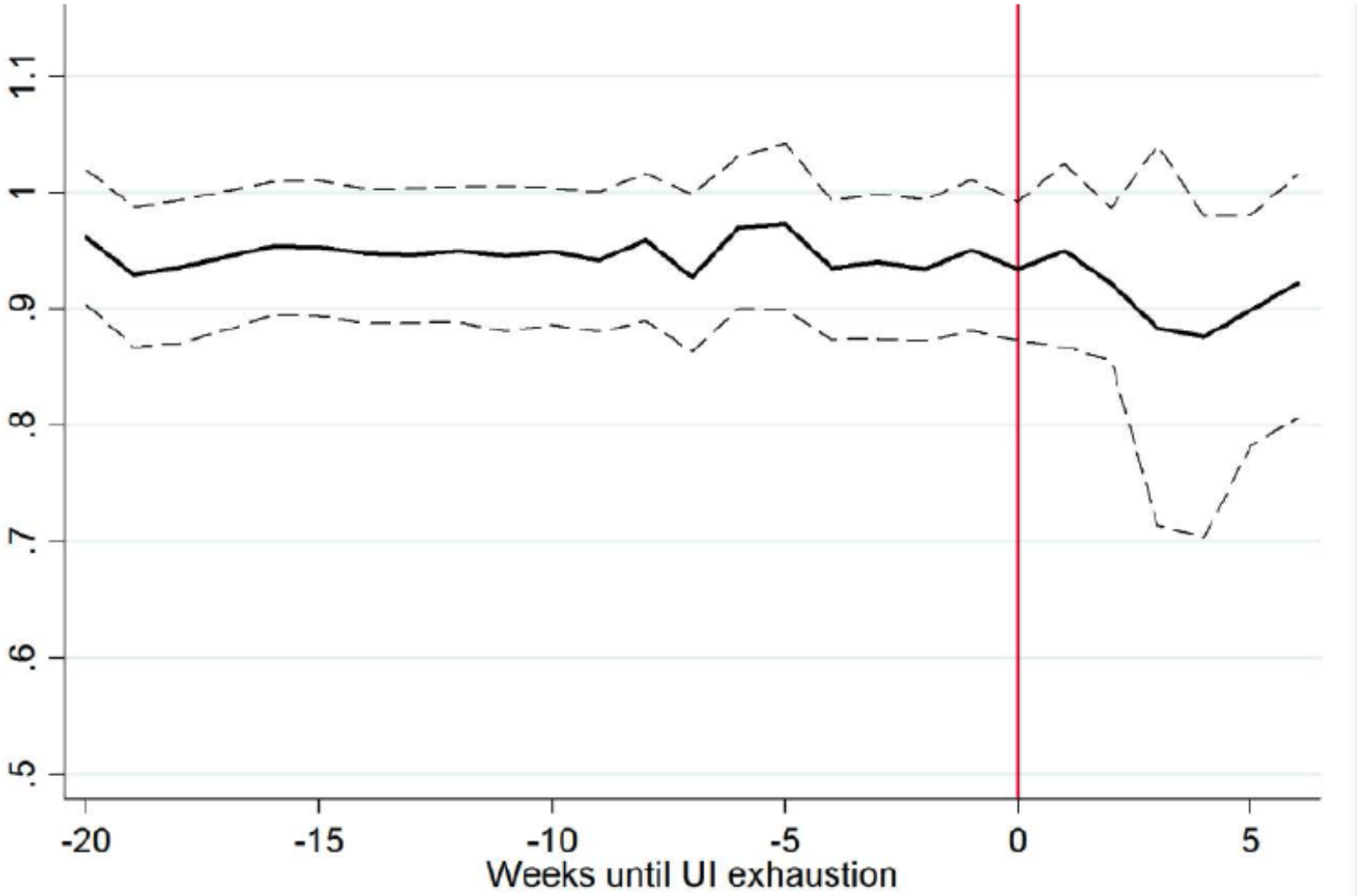
Figure 1: The Reservation Wage over the Spell of Unemployment in the Model



Notes: The figures show the decline of the reservation wage over the spell of unemployment in the calibrated version of the model described in Section 2. The calibration underlying the figure in Panel (a) assumes that consumption after UI benefit exhaustion drops by 16.7 percent compared to consumption before UI exhaustion, whereas the calibration underlying the figure in Panel (b) assumes that the drop is 33.3 percent. The blue dashed line shows the reservation wage path in an environment where unemployment benefits have unlimited duration, whereas the red dashed line shows the reservation wage in an environment with no unemployment

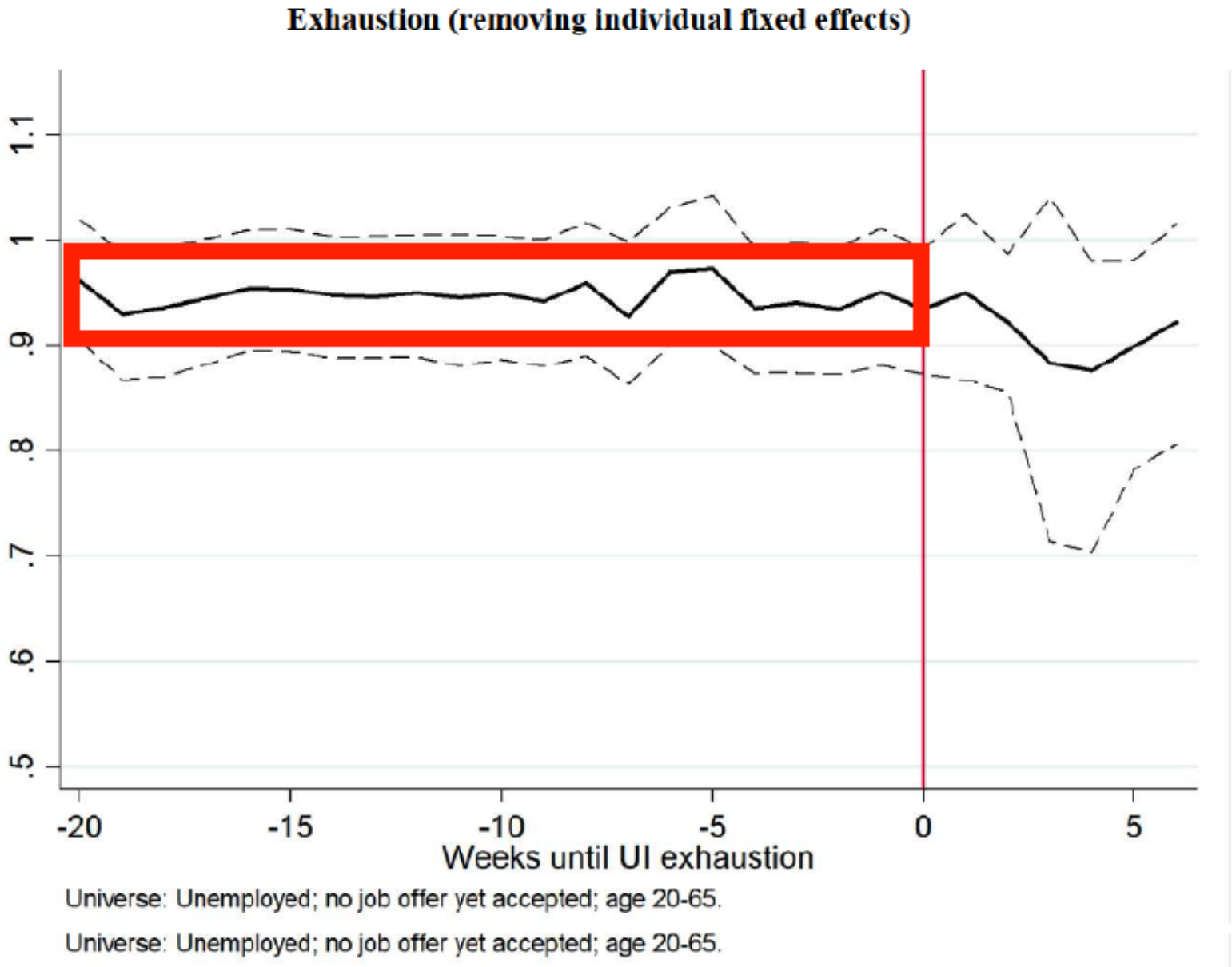
Doesn't seem to work!

Figure 5: Average Ratio of Reservation Wage to Previous Wage, before and after UI Exhaustion (removing individual fixed effects)



Universe: Unemployed; no job offer yet accepted; age 20-65.

Look at the level!



Universe: Unemployed; no job offer yet accepted; age 20-65.

Focusing on the sample of those that received job offers...

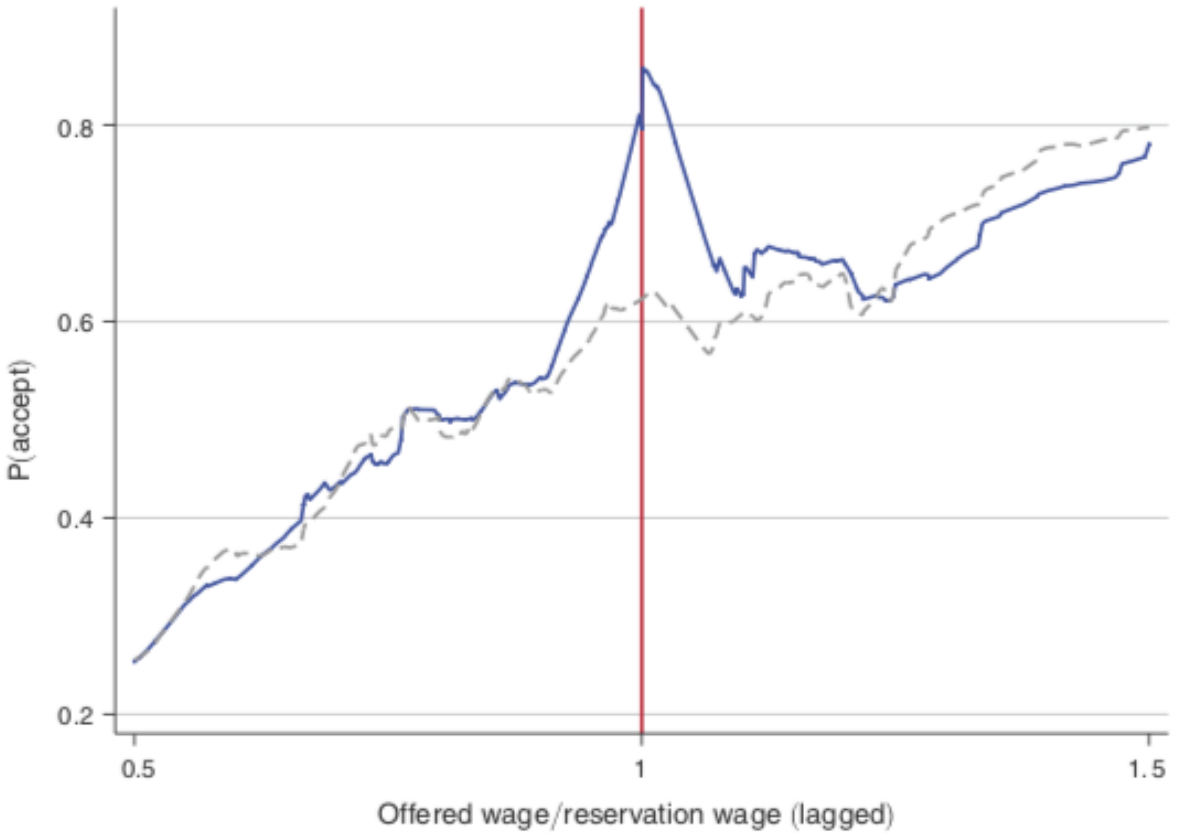


FIGURE 7. LOWESS OF ACCEPTING A JOB OFFER ON THE RATIO OF THE OFFERED WAGE TO THE RESERVATION WAGE

Notes: This graph shows the predicted value of a locally weighted regression (LOWESS) with bandwidth parameter 0.3 for a dummy of whether the job offer was accepted on the ratio of the hourly offered wage to the hourly reservation wage from the previous interview. The dashed line excludes observations where the offered wage equals the reservation wage. Sample: full-time offers, survey respondents aged 20–65.

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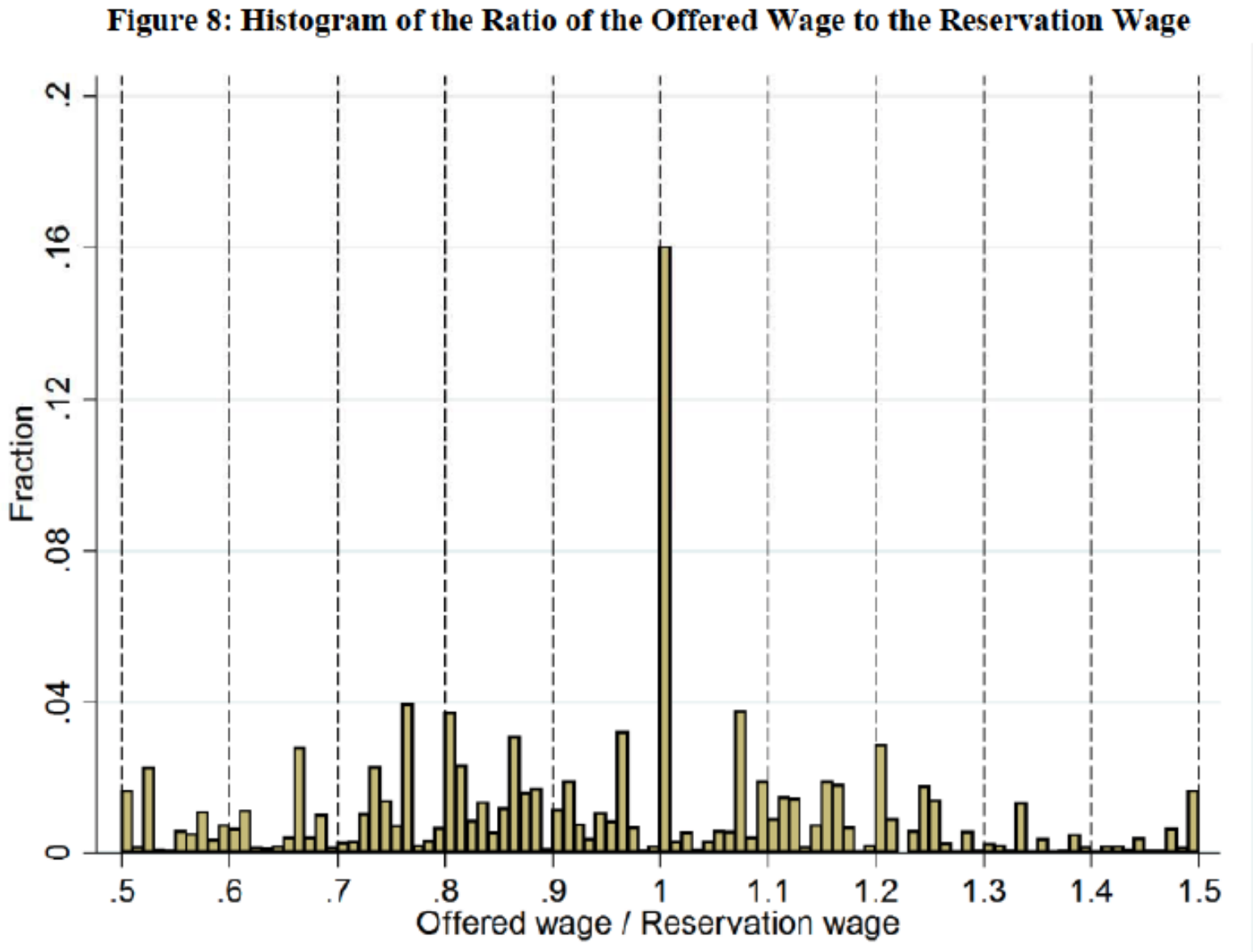
e

Diverse motives for rejecting offers above reservation

Table 8. Reasons for not accepting offer, by wage offer below and above reservation wage

Reason for not accepting offer	Hourly offered wage < hourly reservation wage	Hourly offered wage >= hourly reservation wage	All
Inadequate pay/benefits	40%	1%	30%
Unsuitable working conditions	1%	28%	9%
Would not make use of my experience or skills	14%	4%	11%
Had insufficient experience or skills	1%	4%	2%
Insufficient hours/too many hours	7%	1%	6%
Changed plans	3%	1%	3%
Transportation problem	12%	19%	14%
Better offer	3%	10%	5%
Other	18%	31%	21%
N	121	42	163

What is going on?



Summing up

- Limited impact of UI benefits on reservation wages.

Summing up

- Limited impact of UI benefits on reservation wages.
- But why? Any suggestions?

Section 2

Policy Debate over UI

US employers struggle to find willing workers after pandemic year

Jobless insurance sparks polarising debate as Joe Biden promotes stimulus plans



Matt Glassman has been unable to hire new workers for his Los Angeles restaurant © Carvingblock PR

The inability of companies to attract new workers has sparked a polarising debate over the possible causes, with Republicans and some business figures claiming that overly generous jobless benefits are discouraging people from getting a job.

The chief culprit, they say, is the Biden administration's extension of a $300 per week top-up of unemployment insurance. In higher-paying states, combined benefits can equal up to $600 a week, the equivalent of nearly $16 per hour. That is more than twice the federal minimum wage.

Schmeider, Von Wachter, and Bender (2016)

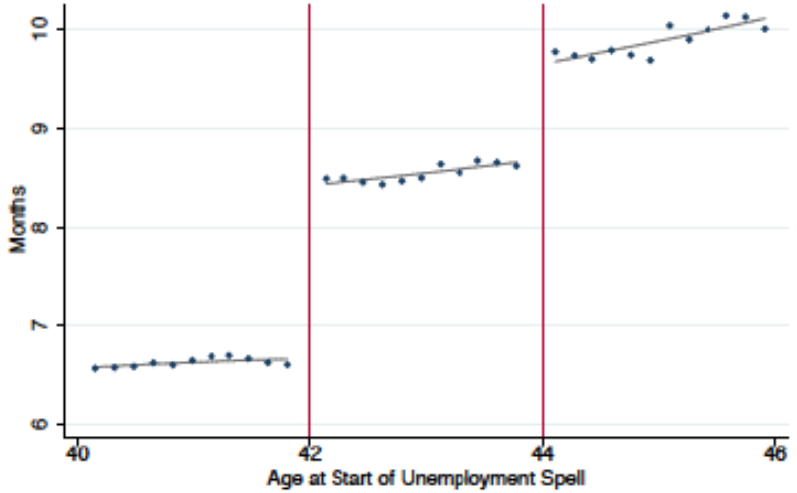
- In standard search models, lengthening/increasing benefit duration should raise reservation wages (thus higher unemployment duration) but also lead to higher re-employment wages
- An earlier experimental literature (Meyer, 1996; Robins and Spiegelman, 2001) finds that hurrying the search process up doesn't do much to the re-employment wage
  - Some of this could just be imprecision
- Recent studies use quasi-experiments with massive datasets to look for a precise effect.
- In Germany, maximum benefit duration increases discontinuously with age, leads to a natural RDD to test for impacts on re-employment wage

Context

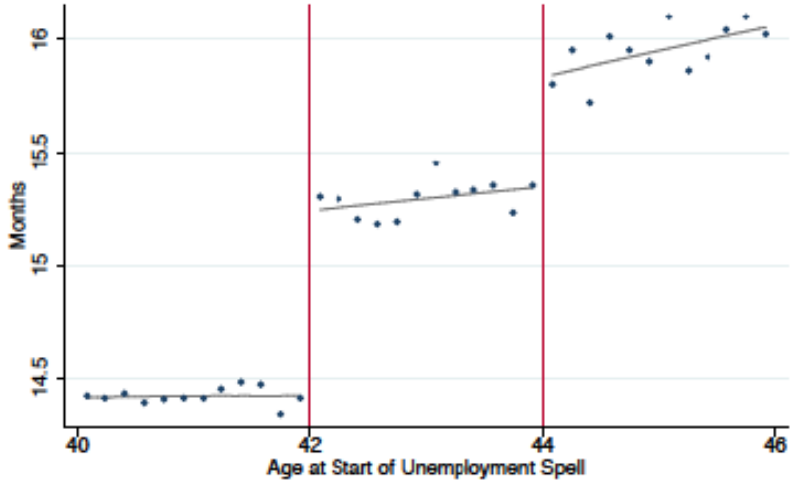
- If involuntary separated from the employer at age 42 or younger: 12 months of maximum coverage of UI ($\approx$ 63% replacement rate if without children, 68% with children)
- If involuntary separated from the employer at age 43 or 44: 18 months of UI.
- At age 45 further increase in maximum duration to 22 months.

Strong first stage effect on durations

Figure 2: The Effect of Extended Potential UI Durations on Benefit and Nonempoyment Durations



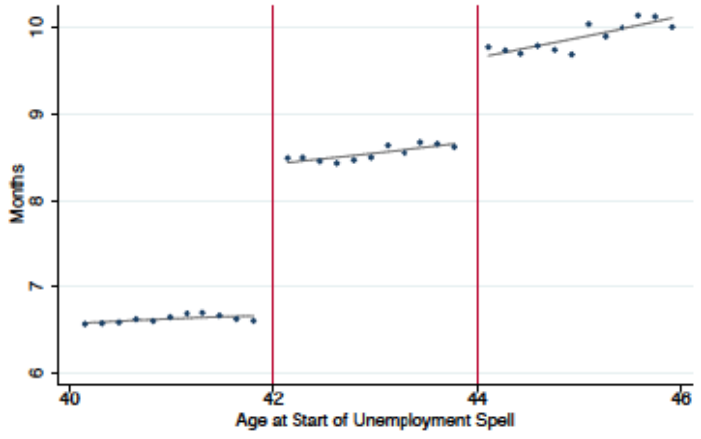
(a) Months of receiving UI benefits



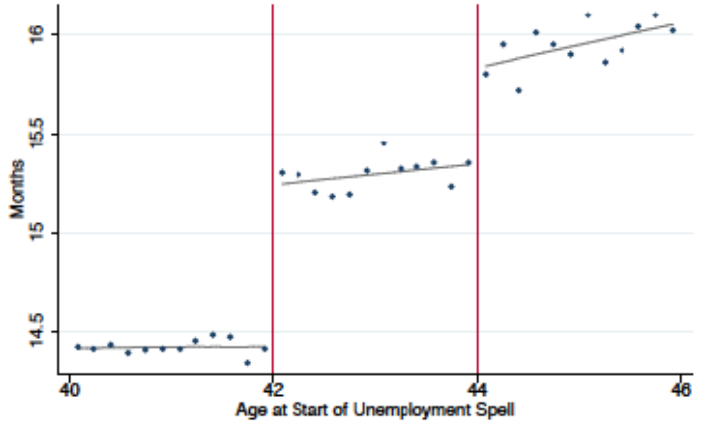
(b) Months of Nonemployment

Extra time = (a) playing Call of Duty or (b) search for better jobs?

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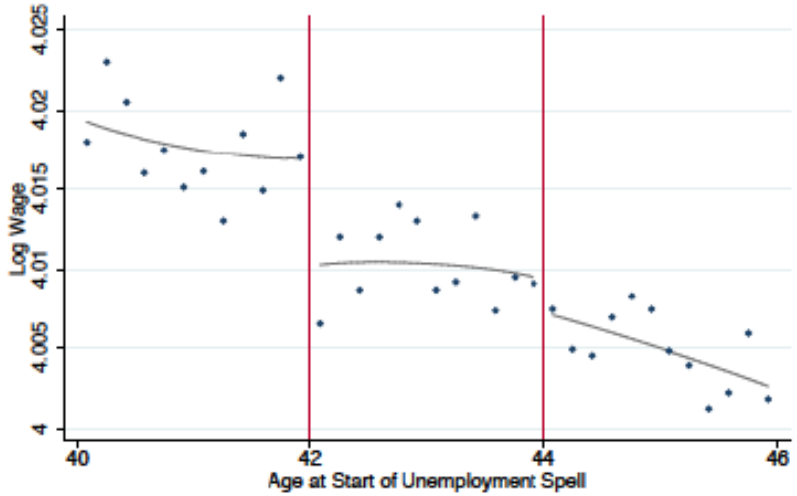
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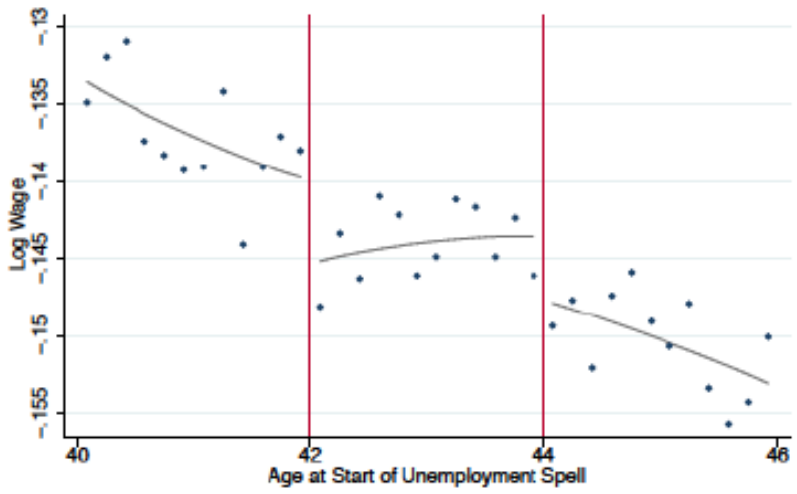
(b) Months of Nonemployment

Small negative impact on re-employment wage!

Figure 3: The Effect of Extended Potential UI Durations on Post Unemployment Wages



(a) Log post unemployment wage



(b) Log wage difference (pre unemployment minus post unemployment)

Making sense of the evidence...

- Negative impact on re-employment wages hard to reconcile with standard search models.
- Note that these models assume that nothing truly happens to the offer distribution if you stay 12 months unemployed as opposed to only 1 month.
- This could unrealistic: skills might decay the more time you spent unemployed. Even if skills don't actually decay, more time spent unemployed can send a negative signal to employers.
- This skill decay / duration dependence argument can reconcile the evidence in Schmeider et al. (2016). But do we have any evidence on this?

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DURATION DEPENDENCE AND LABOR MARKET CONDITIONS: EVIDENCE FROM A FIELD EXPERIMENT*

KORY KROFT
FABIAN LANGE
MATTHEW J. NOTOWIDIGDO

This article studies the role of employer behavior in generating "negative duration dependence"—the adverse effect of a longer unemployment spell—by sending fictitious résumés to real job postings in 100 U.S. cities. Our results indicate that the likelihood of receiving a callback for an interview significantly decreases with the length of a worker's unemployment spell, with the majority of this decline occurring during the first eight months. We explore how this effect varies with local labor market conditions and find that duration dependence is stronger when the local labor market is tighter. This result is consistent with the prediction of a broad class of screening models in which employers use the unemployment spell length as a signal of unobserved productivity and recognize that this signal is less informative in weak labor markets. *JEL* Code: J64.

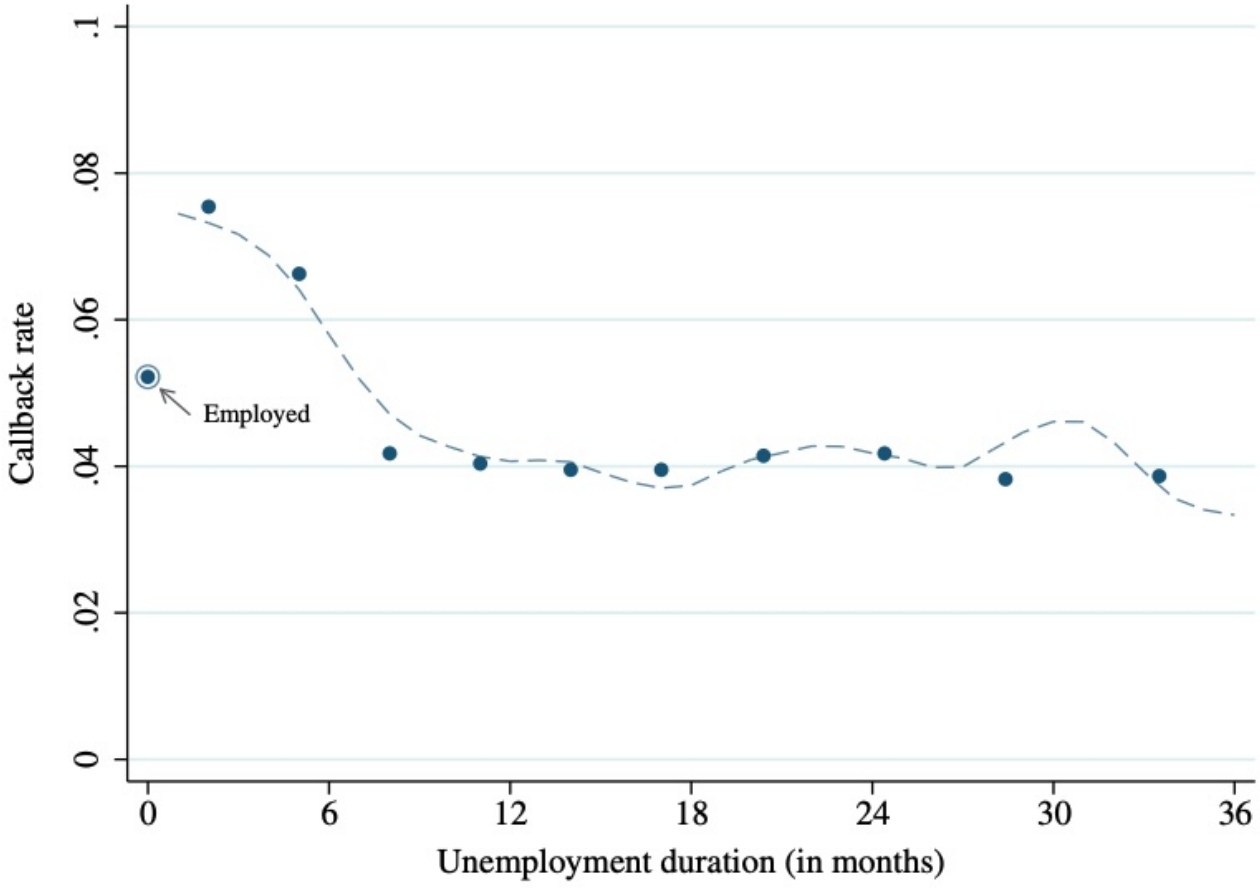


FIGURE II

Callback Rate versus Unemployment Duration

The top figure reports average callback rate by unemployment duration (in months); résumés for which the individual was currently employed are assigned unemployment duration of 0. In the bottom figure, the data are grouped into three- to four-month bins before computing the average callback rate. In both panels, the dashed line is a (smoothed) local mean, which is generated using an Epanechnikov kernel and a bandwidth of two months.

Nekoei and Weber (2017)

- Examine an age-based discontinuity in maximum benefit duration in Austria
- They find that longer durations increase the re-employment wage
- Reconciliation: they argue that duration dependence sometimes outweighs reservation wage effects
- Provide evidence for this argument via meta-analysis of relationship between duration and wage impacts

Small increase in the wage

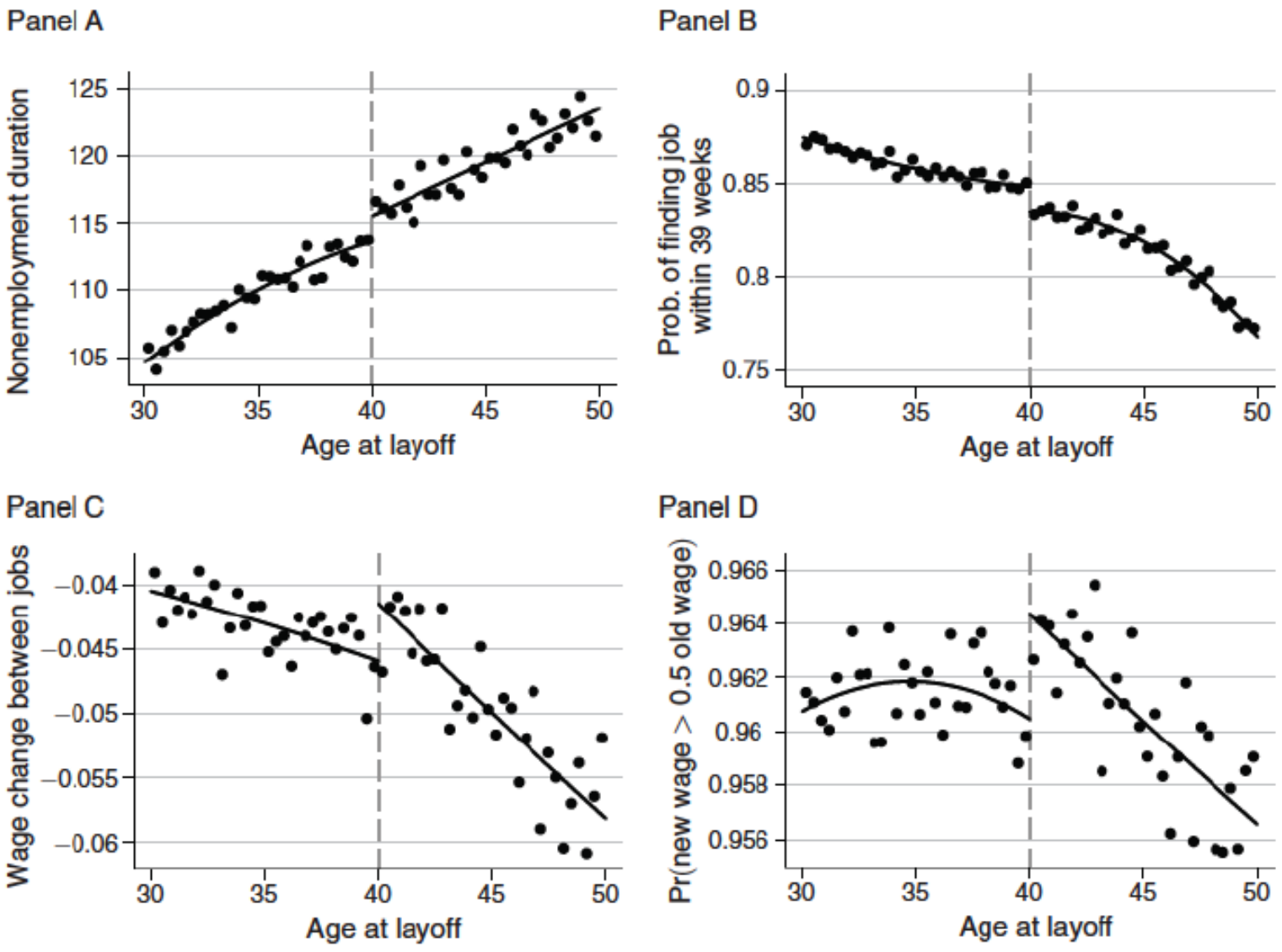
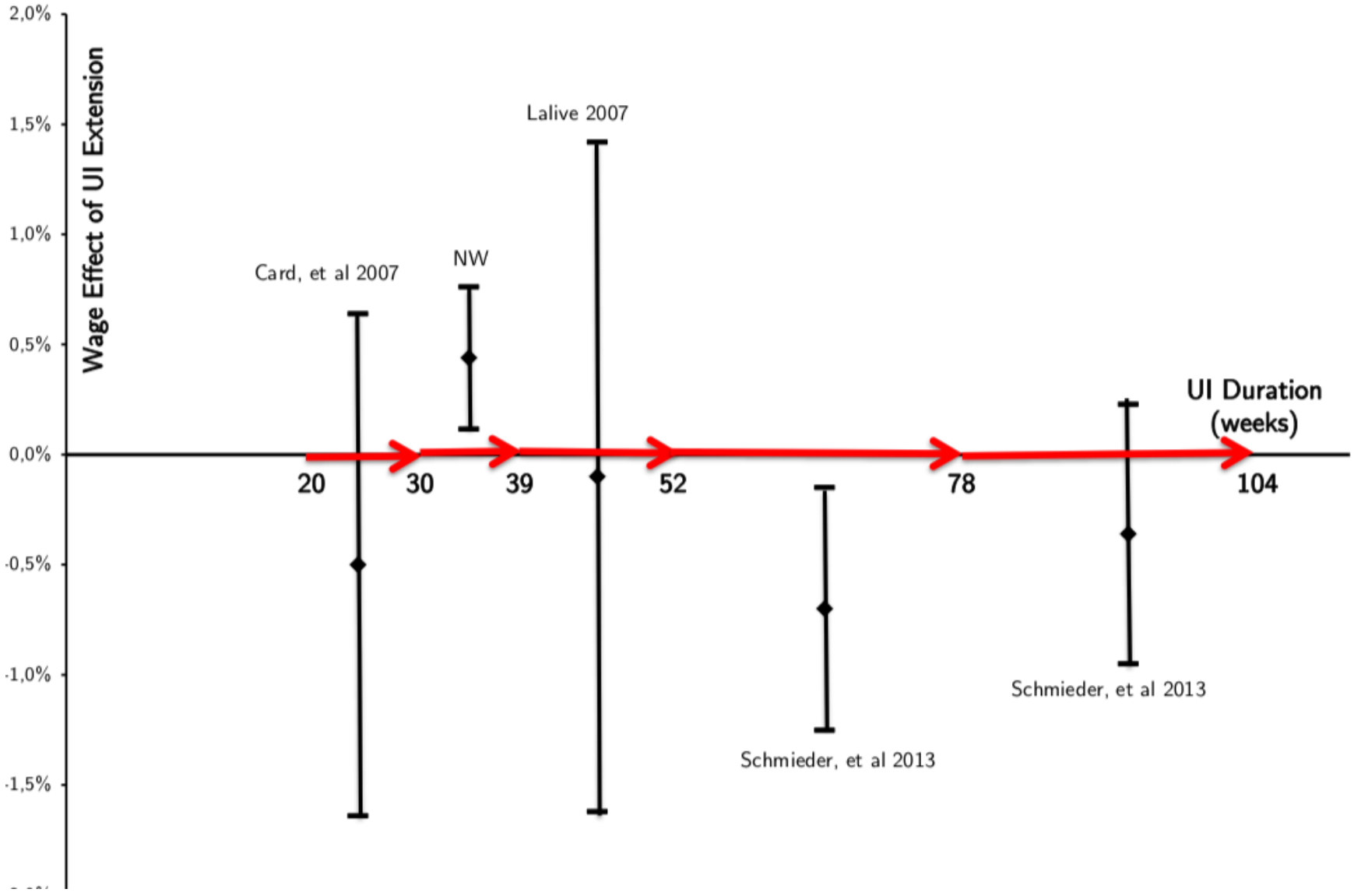
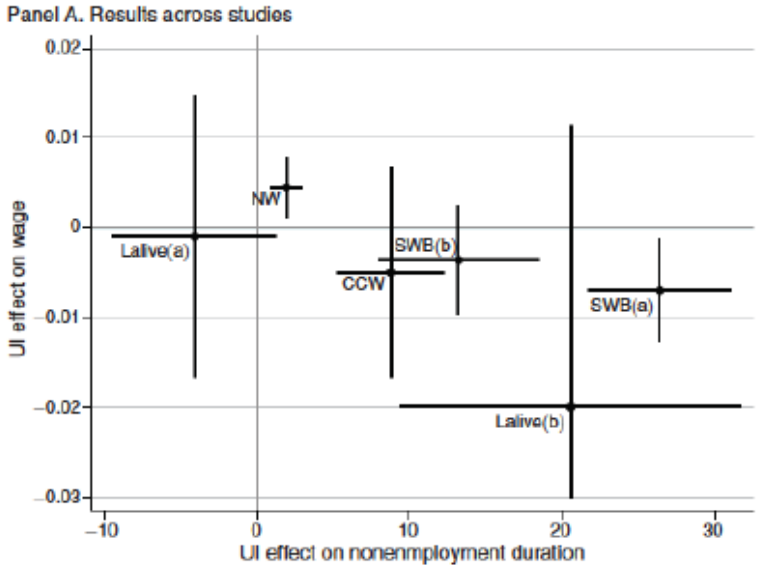


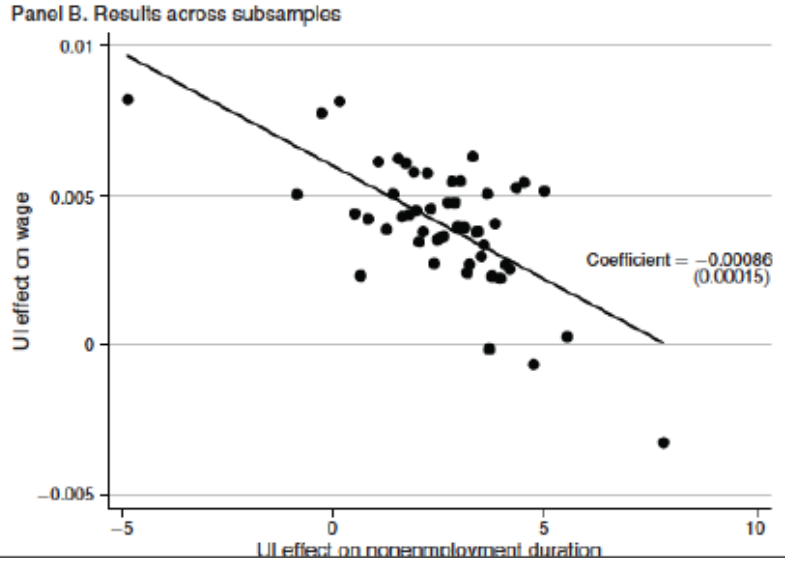
FIGURE 3. UI EFFECTS ON NONEMPLOYMENT DURATION AND WAGE

A meta-analysis



A meta-analysis





Thoughts

- Across all studies of this subject, wage effects are very small (but also first stage)
  - NW: increase in 10 weeks of UI $\rightarrow$ 2 extra days job less (2% increase) $\rightarrow$ 0.5 percent higher wage
- Nekoei and Weber write a directed search model:
  1. UI induces workers to search for better jobs.
  2. But you can also become more picky (search of a perfect job) $\rightarrow$ more unemployed $\rightarrow$ negative duration dependence.
- Heterogeneity in the matching function can generate negative relationship between effect of wages and UI unemployment duration.

More recent Evidence

Early Withdrawal of Pandemic Unemployment Insurance: Effects on Earnings, Employment and Consumption *

Kyle Coombs† Arindrajit Dube‡ Calvin Jahnke§ Raymond Kluender¶
Suresh Naidu‖ Michael Stepner**

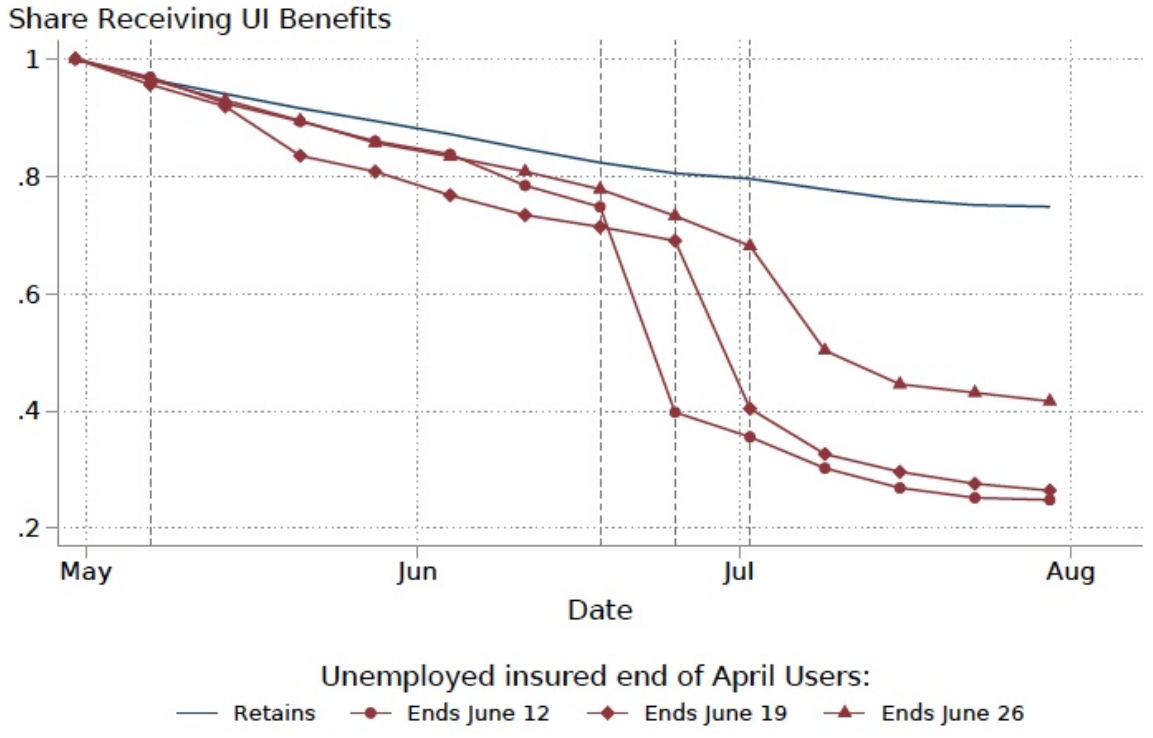
August 20, 2021

Abstract

In June 2021, 22 states ended all supplemental pandemic unemployment insurance (UI) benefits, eliminating benefits entirely for over 2 million workers and reducing benefits by $300 per week for over 1 million workers. Using anonymous bank transaction data and a difference-in-differences research design, we measure the effect of withdrawing pandemic UI on the financial and employment trajectories of unemployed workers in states that withdrew benefits, compared to workers with the same unemployment duration in states that retained these benefits. In our data through August 6, we find that ending pandemic UI increased employment by 4.4 percentage points while reducing UI recipiency by 35 percentage points among workers who were unemployed and receiving UI at the end of April 2021. Through the first week of August, average UI benefits for these workers fell by $278 per week and earnings rose by $14 per week, offsetting only 5% of the loss in income. Spending fell by $145 per week, as the loss of benefits led to a large immediate decline in consumption.

Treated and Control

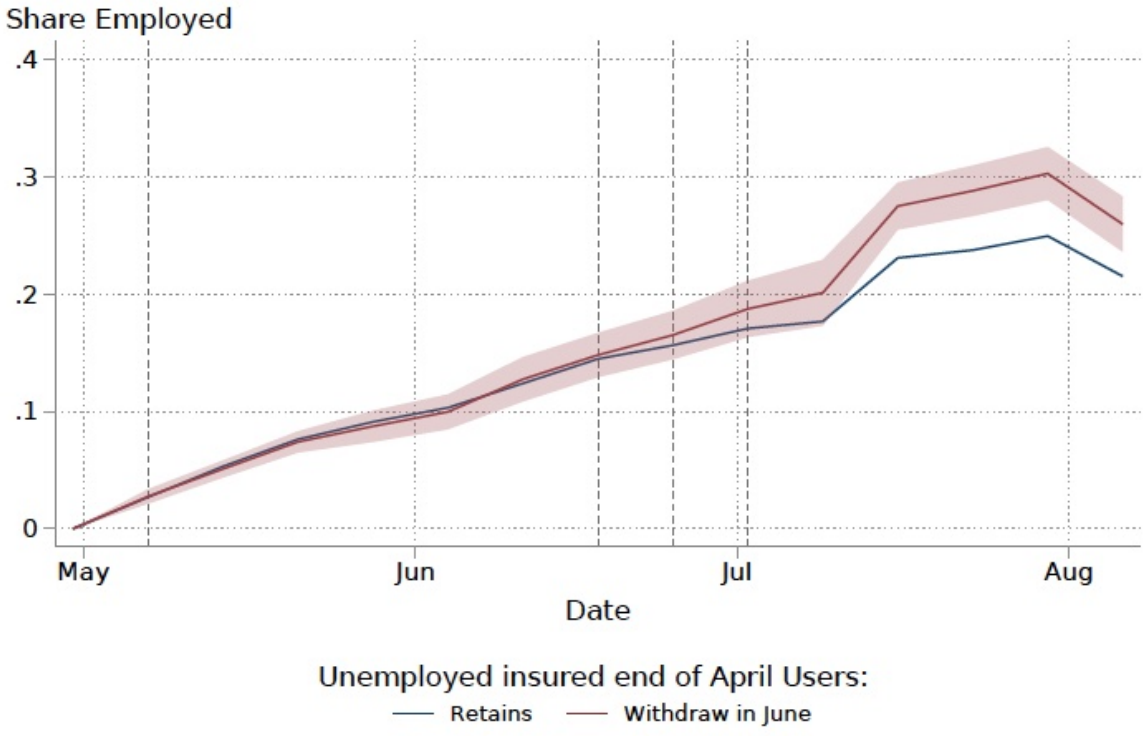
Figure 2. Mean Share Receiving UI Benefits by June Withdrawal State Cohorts



Notes: The above figure plots the mean share of users who are receiving UI benefits in a given week. The sample is restricted to those 18,648 Earnin users whom we track as receiving UI benefits and no earnings in the final week of April, resulting in the first observation being mechanically 1 for all states. In our sample, the Retain cohort contains 23 states, June 12

Effects on Employment

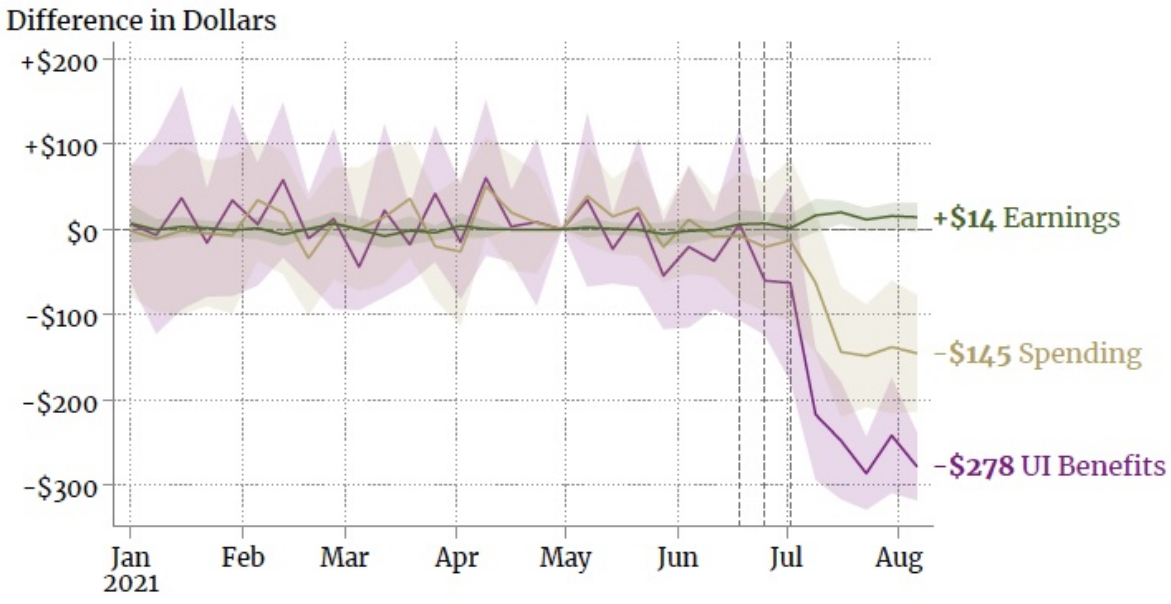
Figure 3. Mean Share Employed by June UI Withdrawal



Notes: The above figure plots the mean share of users who are employed in a given week. The sample is restricted to those 18,648 Earnin users whom we track as receiving UI benefits and no earnings in the final week of April, resulting in the first observation being mechanically 0 for all states. In our sample, the Retain cohort contains 23 states and June Withdrawal contains 19.

An awesome figure

Figure 5. Difference-in-differences Estimates: Effect of June UI Withdrawal on UI Inflows, Earnings, and Spending Levels

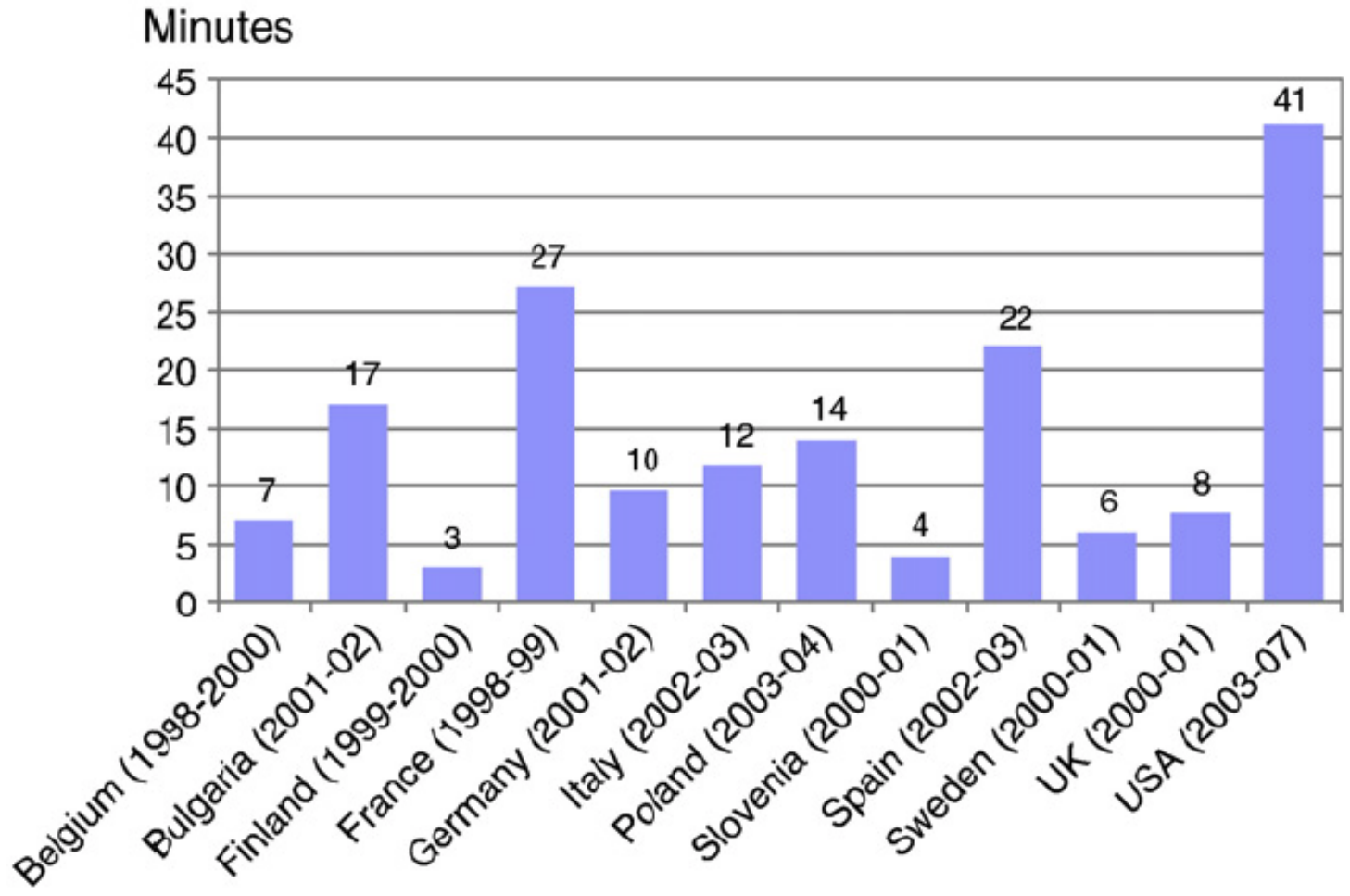


Notes: The above figure plots the difference in means between June Withdrawal and Retain states for three separate dollar measures: UI Inflows, Spening, and Earnings. The sample is restricted to those 18,648 Earnin users whom we track as receiving UI benefits and no earnings in the final week of April. Values for the last week of April are normalized to zero. In our sample, the Retain cohort contains 23 states and June Withdrawal contains 19.

Krueger and Mueller (2010)

- What does it mean to be unemployed? How can we measure search intensity?
- Official definition: "people are classified as unemployed if they do not have a job, have actively looked for work in the prior 4 weeks, and are currently available for work."
- Krueger and Mueller study to see what nominally unemployed individuals are doing with their time.
- Data from ATUS from 2003-2007.

Job search does not seem time intensive



Notes : See Krueger and Mueller (2008a) for details about the underlying time use data.

Fig. 1. Average number of minutes devoted to job search per day on weekdays by unemployed workers in various countries.

Unemployed search more than the employed

Table 1a
Descriptive statistics ATUS 2003–2007, by labor force status (weekdays and weekends).

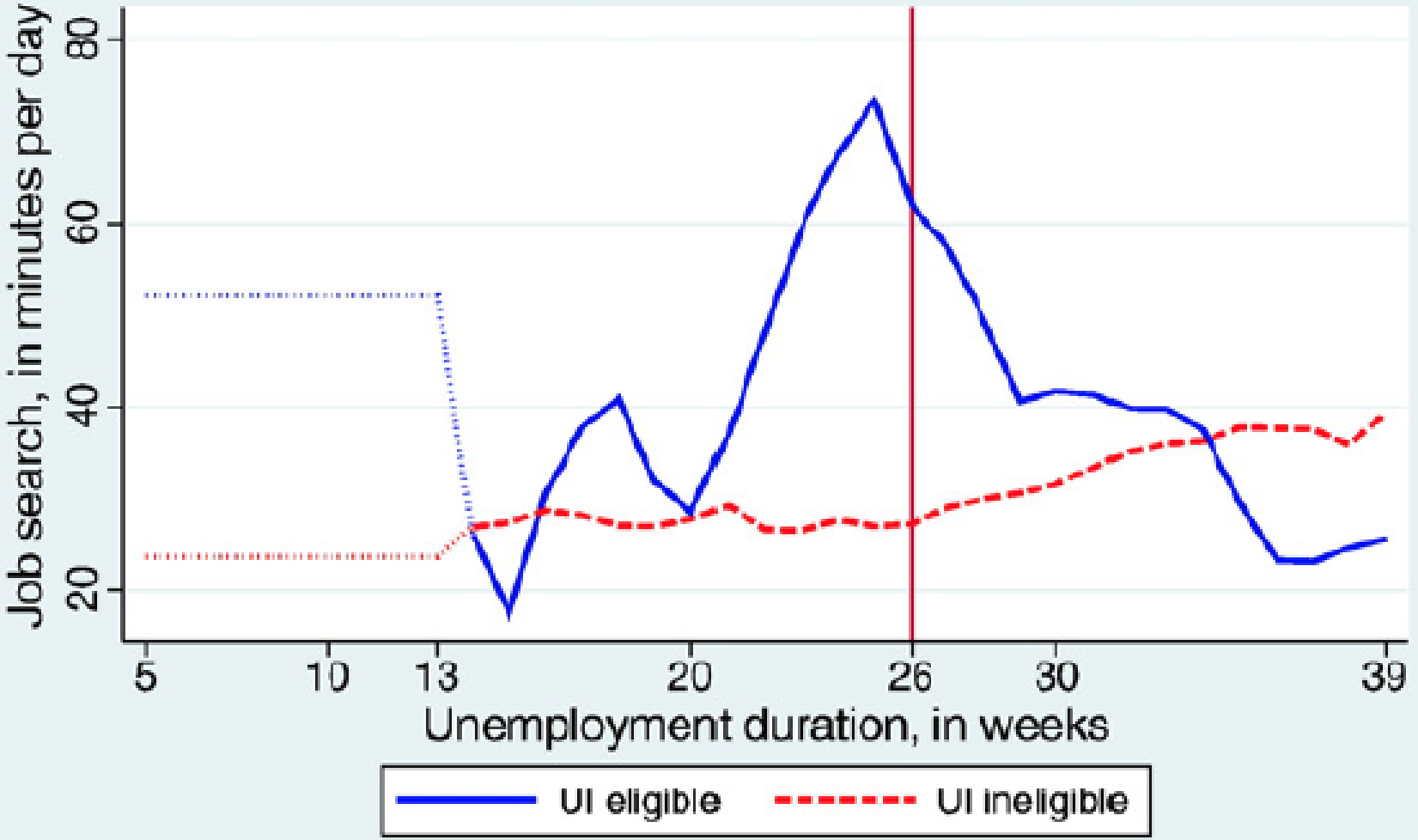
	# of respondents	% of total	Average job search, in min per day	Participation in job search	Average job search (participants), in min per day
By labor force status					
Employed	42,934	76.4%	0.6	0.6%	101.0
Unemployed	2171	3.9%	32.1	19.3%	166.9
Not in labor force	11,091	19.7%	0.8	0.5%	152.9
By type of employed		(% of employed)			
Working in CPS	40,576	94.5%	0.5	0.5%	107.6
Unemployed in CPS	824	1.9%	2.8	2.5%	115.4
Not in labor force in CPS	1534	3.6%	0.8	1.7%	49.7
By type of unemployed		(% of unemployed)			
Jobloser	943	43.4%	45.2	27.5%	164.2
On temporary layoff w/ recall expectation	344	15.8%	13.2	7.1%	185.8
Jobleaver	65	3.0%	52.9	24.9%	212.3
Re- or new entrant	819	37.7%	23.1	14.1%	163.6
By UI eligibility status		(% of unemployed)			
UI ineligible	1000	46.1%	25.4	15.6%	163.4
UI eligible	1171	53.9%	38.0	22.5%	169.1
By type of "not in labor force"		(% of not in labor force)			
Working in CPS	1181	10.6%	2.4	1.8%	134.1
Unemployed in CPS	305	2.7%	4.2	3.2%	130.8
Not in labor force in CPS	9605	86.6%	0.5	0.3%	176.7

Notes: Averages and participation rates are computed with survey weights. Both weekdays and weekends are included in the sample. Universe: Civilian, non-institution population, aged 20–65.

Benefits strongly correlated with time use

Dependent variable: time allocated to job search, in min per day	Mean (Std)	Full sample (1)	Subsample (2): eligible w/o recall expectation and unempl. dur.<=26	Subsample (3): ineligible	Wage equation-dependent variable: log(hourly wage)
Mean of dependent variable		32.1	49.1	25.4	2.76
Log(maximum weekly benefit amount)	5.89	−6.86	−57.275	10.096	
	(0.220)	(11.971)	(30.663)*	(19.864)	
Fitted log(hourly wage)	2.60	110.066	174.048	105.099	
	(0.329)	(48.715)**	(120.772)	(64.247)	
Std(residual of wage equation) — by state	0.490	92.868	274.379	83.161	
	(0.023)	(101.732)	(196.089)	(111.950)	
On temporary layoff w/ recall expectation (1)	0.15	−32.884		−11.497	
		(4.973)***		(12.479)	
Jobleaver	0.03	12.876		21.507	
		(16.585)		(20.857)	
Re- or new entrant	0.38	−13.656		−3.456	
		(5.280)**		(10.363)	
Age	36.75	−5.12	−6.816	−5.605	0.061
		(3.198)	(7.966)	(3.691)	(0.001)***
Age^2		0.053	0.078	0.052	−0.001
		(0.034)	(0.086)	(0.039)	(0.000)***
Some college or associate degree (2)	0.29	−13.133	−16.282	−14.284	0.209
		(12.991)	(32.615)	(14.991)	(0.002)***
College degree (BA, MA or PhD)	0.16	−46.877	−59.764	−68.348	0.573
		(28.113)	(72.634)	(37.407)*	(0.003)***
Female	0.51	14.021	52.805	−6.649	−0.231
		(13.543)	(33.296)	(16.080)	(0.002)***
Female*partner	0.28	−11.09	−34.334	9.703	
		(8.400)	(16.167)**	(17.016)	
Female*children	0.30	−7.925	−26.06	6.872	
		(14.362)	(26.744)	(17.905)	
Partner	0.50	0.176	7.652	−11.682	
		(8.911)	(13.632)	(18.347)	
Children	0.49	7.113	39.751	−14.717	
		(12.786)	(18.914)**	(17.389)	
Weekend	0.28	−30.883	−53.138	−21.693	
		(3.797)***	(6.492)***	(4.676)***	
Constant		−115.341	−71.375	−169.555	1.2
		(66.062)*	(128.577)	(100.181)*	(0.013)***
Year and month dummies		x	x	x	Year dummy
State dummies					x
Observations		2171	671	1000	319,813
R-squared		0.09	0.16	0.13	0.29

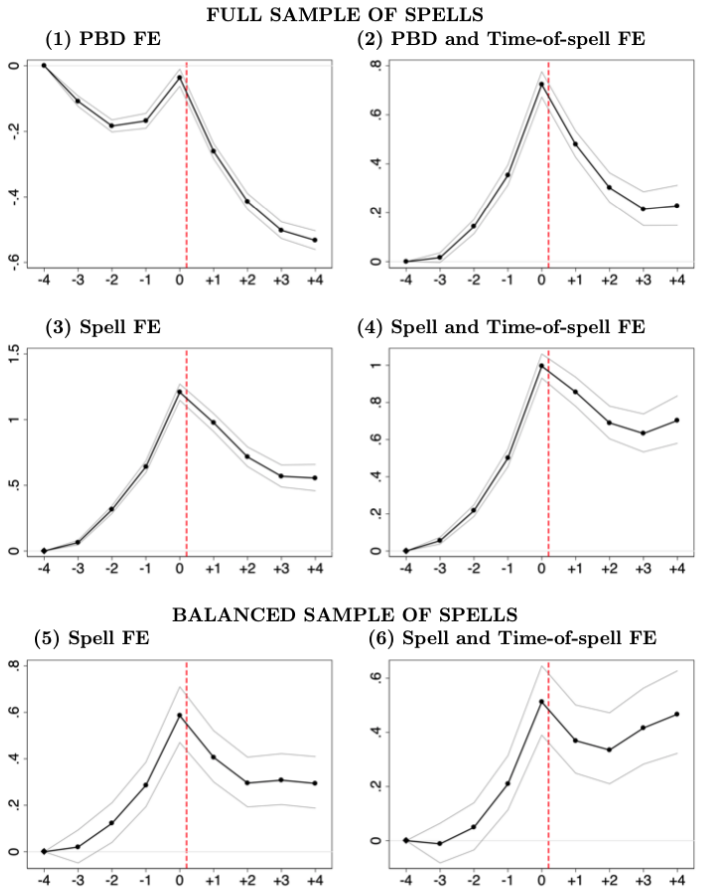
Procrastination?



Notes: Bandwidth = 0.1. Survey weights are used to compute the lowess smoother.
Unemployed with an expectation of recall to their previous employer are excluded from the sample.
The dotted lines refer to the average of time spent on job search before week 14.

2020 Version of this plot: Marinescu and Skandalis (2020)

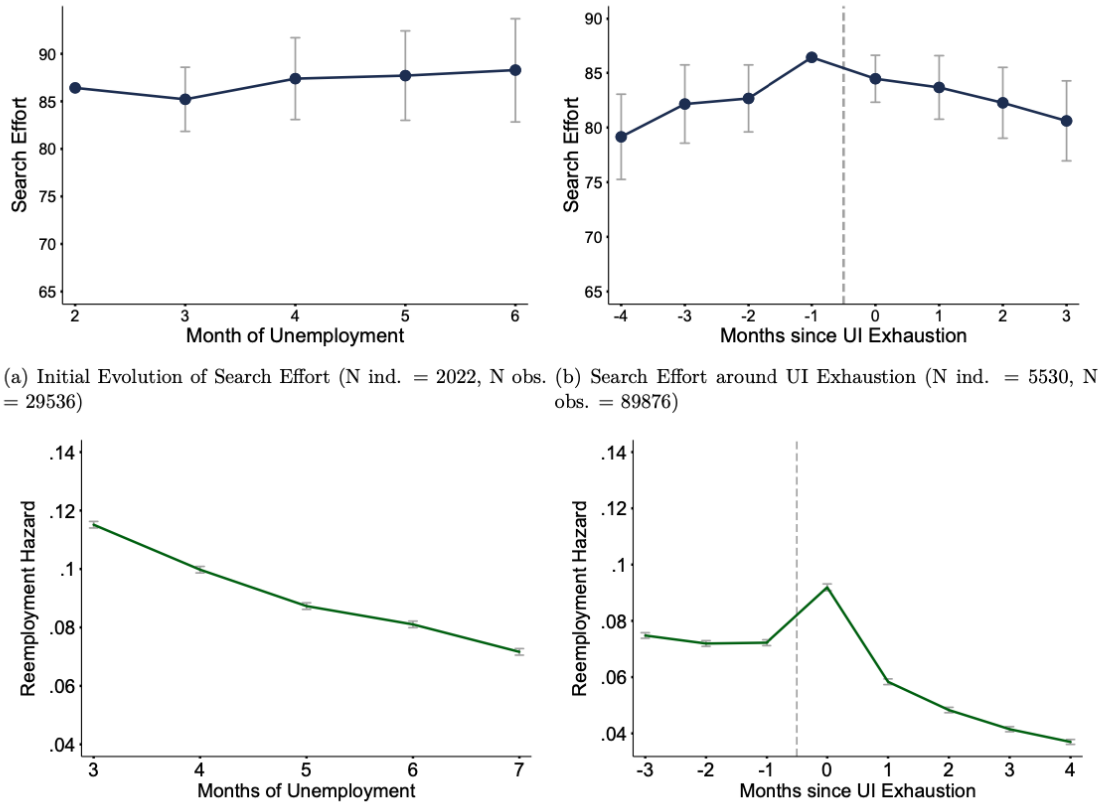
Figure IV:
The search effort of unemployed workers around benefits exhaustion



Notes: The graphs present the estimates from Poisson regressions of search effort on time-to-benefit-exhaustion, reported Table III. The x-axis denotes quarters relative to benefit exhaustion, the y-axis denotes search effort level relative to 4 quarters before benefit exhaustion (note that the scales of the y-axis are different from one panel to the other for clarity). In panel (1), we include PBD-group FE; in (2), we add time-of-spell FE; in (3), we include spell FE; in (4), we present our preferred specification, with both time-of-spell and spell FE. (5)–(6) correspond to the same specifications as (3)–(4), but we use the balanced sample of spells (defined in section 5.1). The grey lines denote 95% confidence intervals based on standard errors clustered at the spell level.

2020 Version of this plot: DellaVigna, Heining, Schmieder and Trenkle (2020)

Figure 6: Search Effort Throughout the Unemployment Spell



(a) Initial Evolution of Search Effort (N ind. = 2022, N obs. = 29536)

(b) Search Effort around UI Exhaustion (N ind. = 5530, N obs. = 89876)

(c) Initial Evolution of Hazard Rate

(d) Hazard Rate around UI Exhaustion

Notes: The figure shows mean job search over the initial spell of unemployment (up to 6 months) and around UI-exhaustion (between -4 and +3 months around UI exhaustion) controlling for individual, weekdate and calender-month fixed effects and compares it to reemployment hazard in those months. For the initial evolution of Search Effort only individuals with $P \geq 8$ are included. Standard Errors are clustered on the Person level. Hazard rates are pooled over different P-groups where each group is weighted with the number of individuals that are in the respective survey group.

Unemployment and Well-Being

Table 4. Self-Reported Health Status for Workers Age 25–54 by Labor Force Status[a]

	Labor force status (percent)		
Health status	*Employed*	*Unemployed*	*Not in the labor force*
Men			
Excellent	20.0	19.5	12.3
Very good	36.3	29.2	20.6
Good	31.9	35.1	24.4
Fair	10.7	13.9	25.4
Poor	1.2	2.3	17.3
No. of observations	7,277	468	683
Women			
Excellent	20.9	16.3	16.6
Very good	37.0	25.6	24.0
Good	30.9	36.3	28.0
Fair	10.0	18.1	19.3
Poor	1.1	3.7	12.1
No. of observations	7,453	637	2,265

Sources: U.S. Bureau of Labor Statistics, American Time Use Survey, Well-Being Module; author's calculations.

a. The sample is pooled over 2010, 2012, and 2013 for individuals age 25–54. The data are weighted using the Well-Being Module's final weights.

Section 3

The Consequences of losing a job

Now let's look at things from a different angle

- What are the consequences of a job loss for an individual perspective?
- Neo-classical theory: I will always get paid my marginal product of labor.
- Things are of course much more complicated (lost of firm specific skills, stigma, rents, etc).

Jacobson, LaLonde and Sullivan (1993)

- Jacobson, LaLonde and Sullivan (1993) look at the effects of job displacement on workers' earnings trajectories
- JLS use matched administrative records on workers and firms in Pennsylvania to look at what happens when high-tenure (6 or more years) workers lose their jobs
- Worker data: Unemployment insurance records
- Firm data: ES202 forms (filed by all firms covered by UI)
- To focus on involuntary job separations, construct a "mass-layoff" sample of separations that occured at firms during times when employment dropped by 30% or more

Jacobson, LaLonde and Sullivan (1993)

- JLS equation:

$$y_{it} = \alpha_i + \gamma_t + x'_{it}\beta + \sum_{k \geq -m}^{M} D_{it}^k \delta_k + \epsilon_{it}$$

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$$y_{it} = \alpha_i + \gamma_t + x_{it}'\beta + \sum_{k \geq -m}^{M} D_{it}^k \delta_k + \epsilon_{it}.$$

- Let t_i^* denote the year of displacement of individual i.
- D_{it}^k: one if individual i was displaced exactly k periods before t

$$D_{it}^k = \mathbf{1}\{t = t_i^* + k\} \tag{8}$$

- Since some k's are negative, this specification also looks at earnings dynamics before separation $\rightarrow$ pre-trends
- Worker and time effects $\implies$ this is a differences in differences estimator

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- Since some k's are negative, this specification also looks at earnings dynamics before separation $\rightarrow$ pre-trends
- Worker and time effects $\implies$ this is a differences in differences estimator
- Who is the control group here?
- δ_k coefficients compare changes in earnings for workers whose displacement status changed to changes for workers who were not displaced at the same time

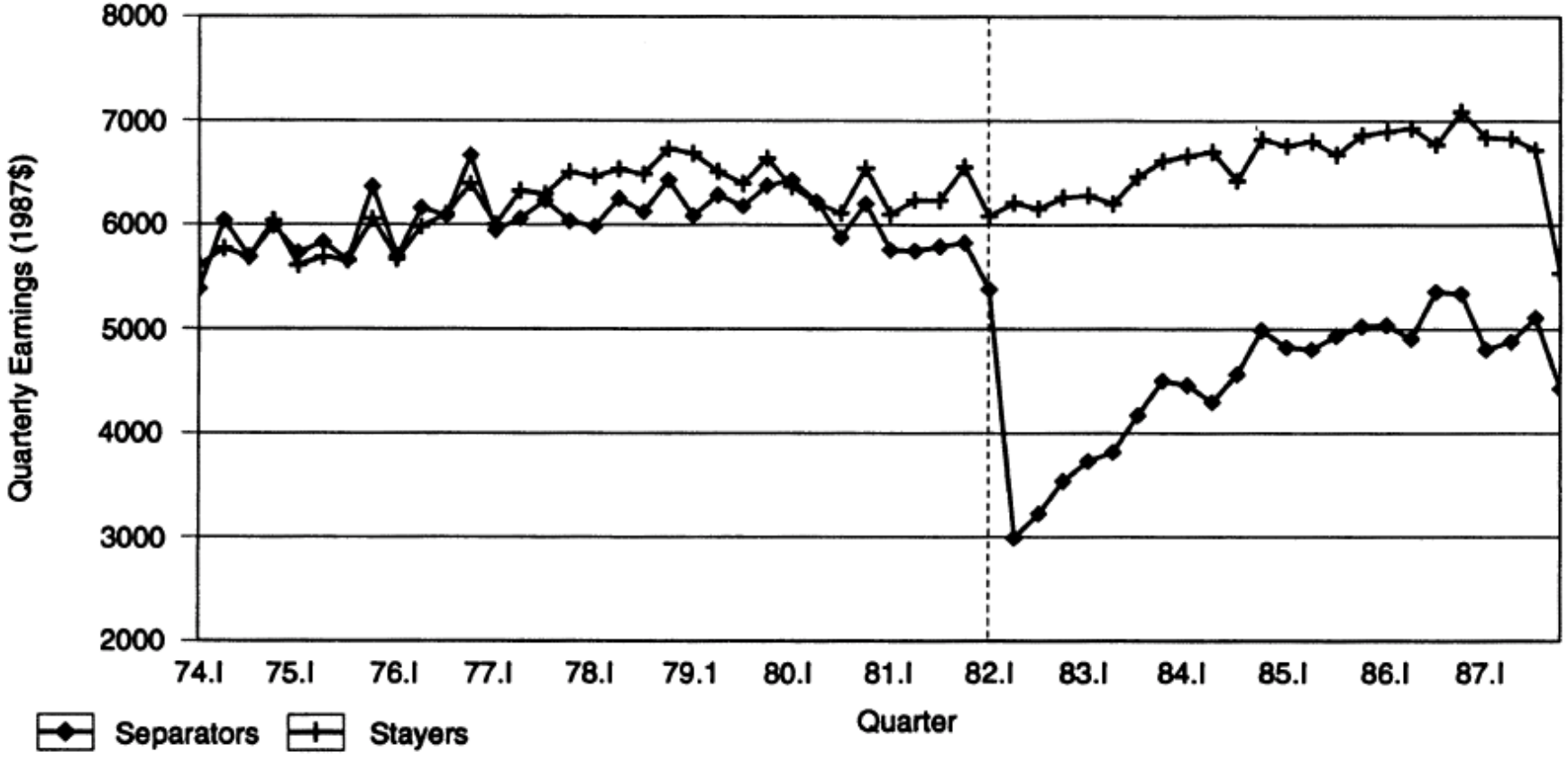


FIGURE 1. QUARTERLY EARNINGS (1987 DOLLARS) OF HIGH-ATTACHMENT WORKERS SEPARATING IN QUARTER 1982:1 AND WORKERS STAYING THROUGH QUARTER 1986:4

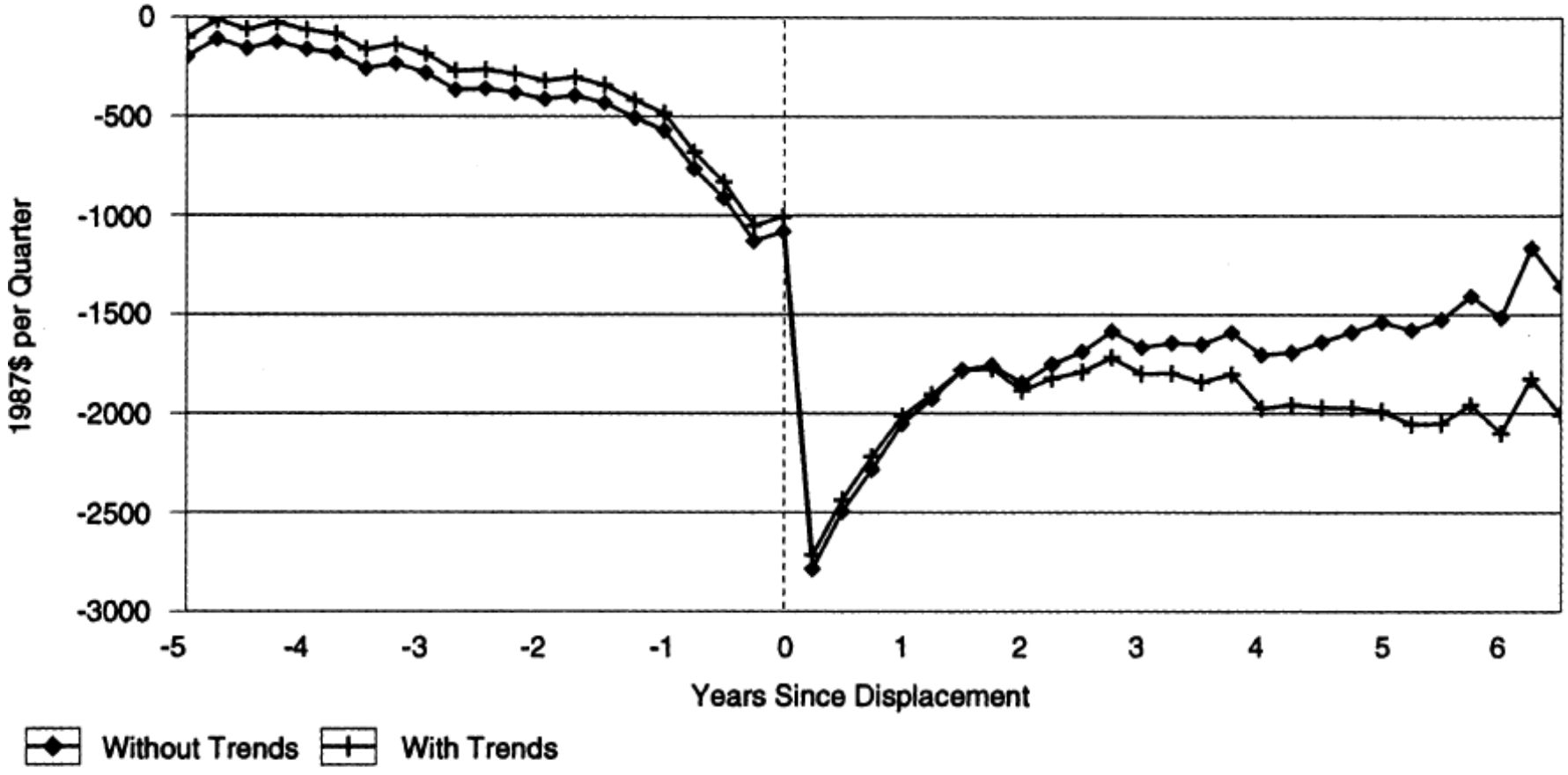


FIGURE 2. EARNINGS LOSSES FOR SEPARATORS IN MASS-LAYOFF SAMPLE

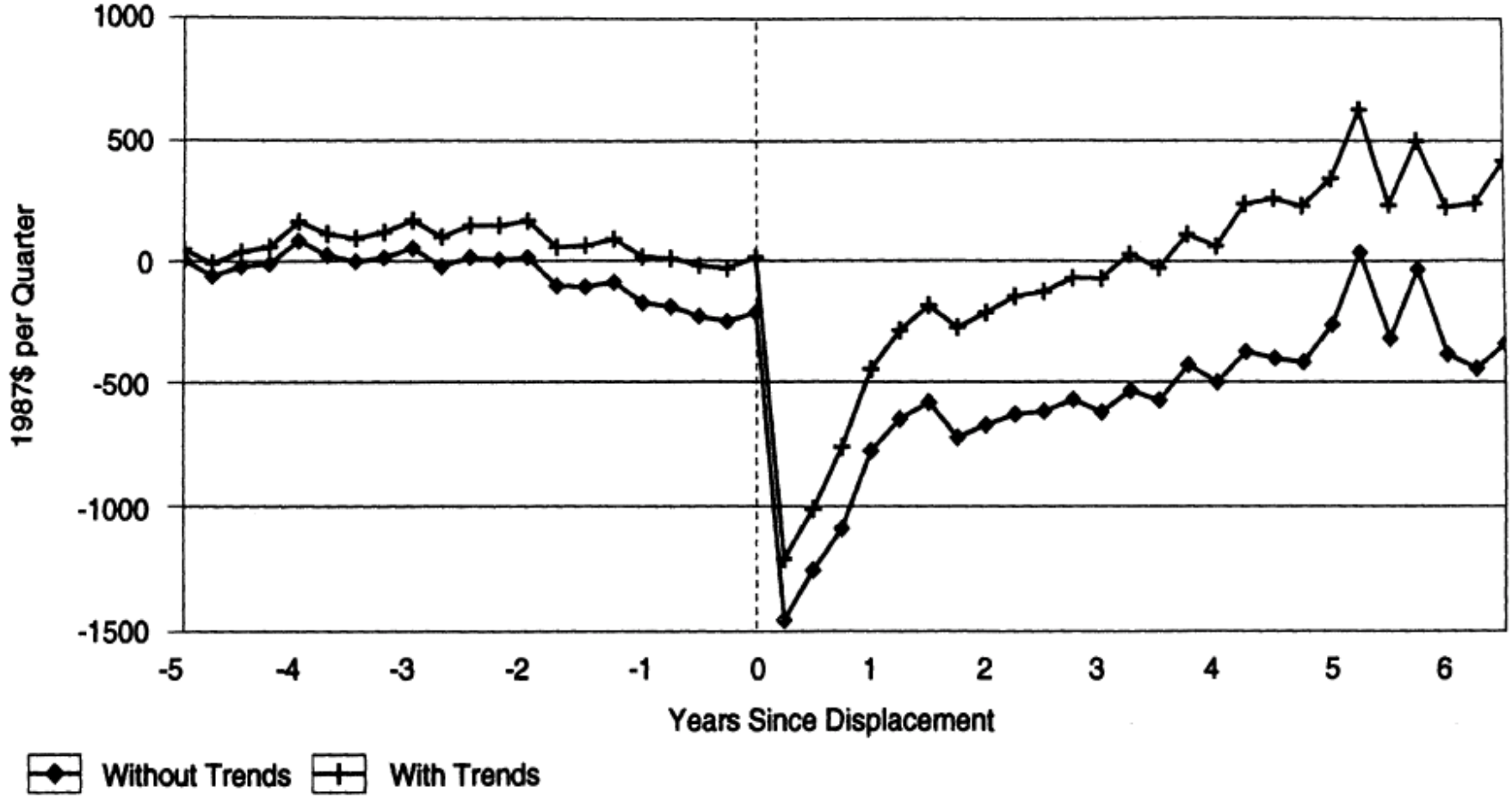


FIGURE 3. EARNINGS LOSSES FOR SEPARATORS IN NON-MASS-LAYOFF SAMPLE

Jacobson, LaLonde and Sullivan (1993)

- JLS findings:
  - Relative earnings for workers who will experience mass layoffs start to decline in advance, then drop sharply at displacement
  - Quick partial recovery, but stabilize around 25% below initial level
  - Younger workers experience larger losses, but faster recovery
  - Larger losses in mining/construction/public utilities; perhaps loss of high-rent union jobs?
  - Long-term losses even for workers reemployed in same industry. JLS interpret this as evidence of firm-specific skills...

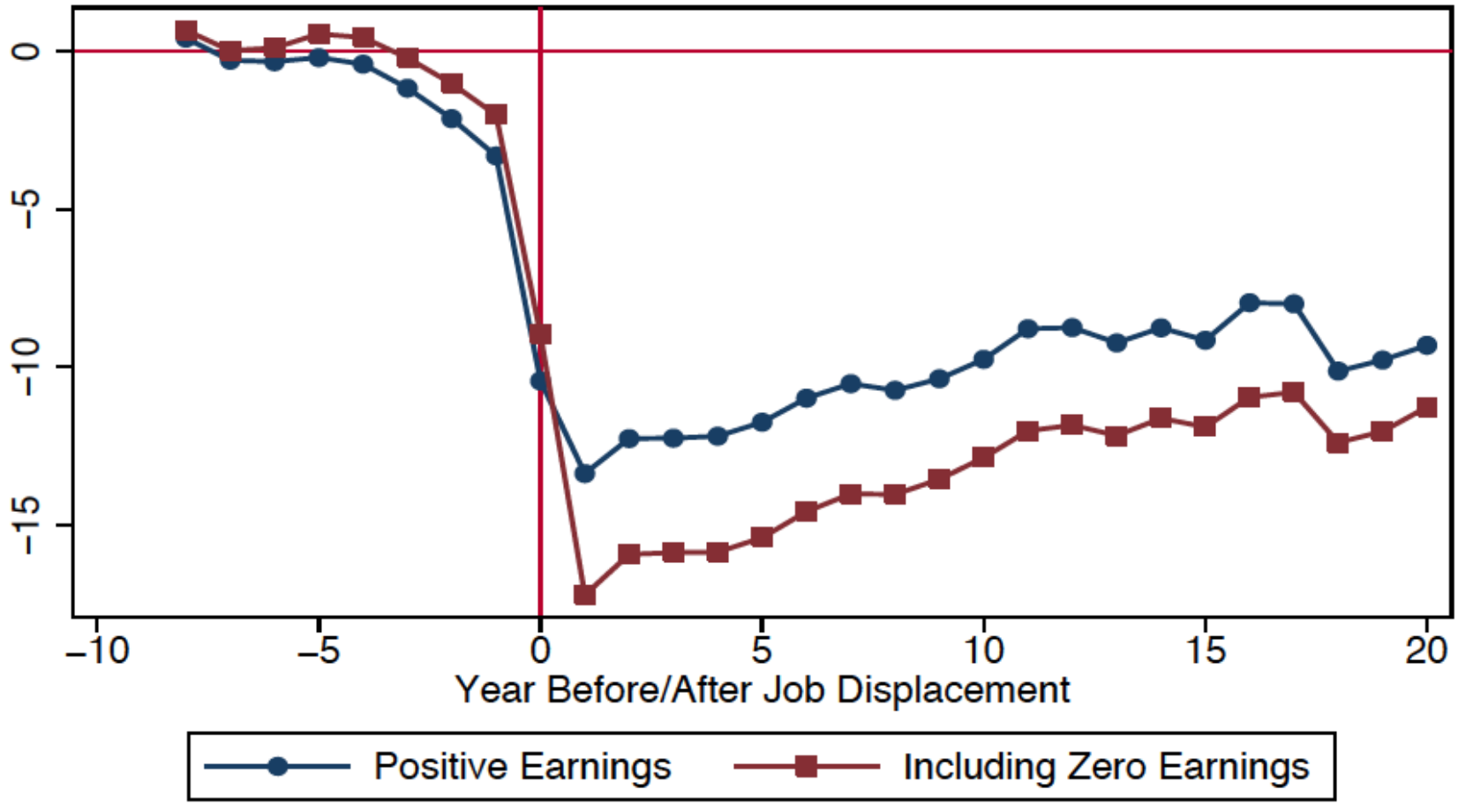
Praise of JLS

- Incredibly important paper for empirical labor economics
  1. Empirical tool (Event study design).
  2. Importance of the question.
  3. Use of high quality administrative data.
  4. Transparent design $\rightarrow$ replication in other countries and for different outcomes.

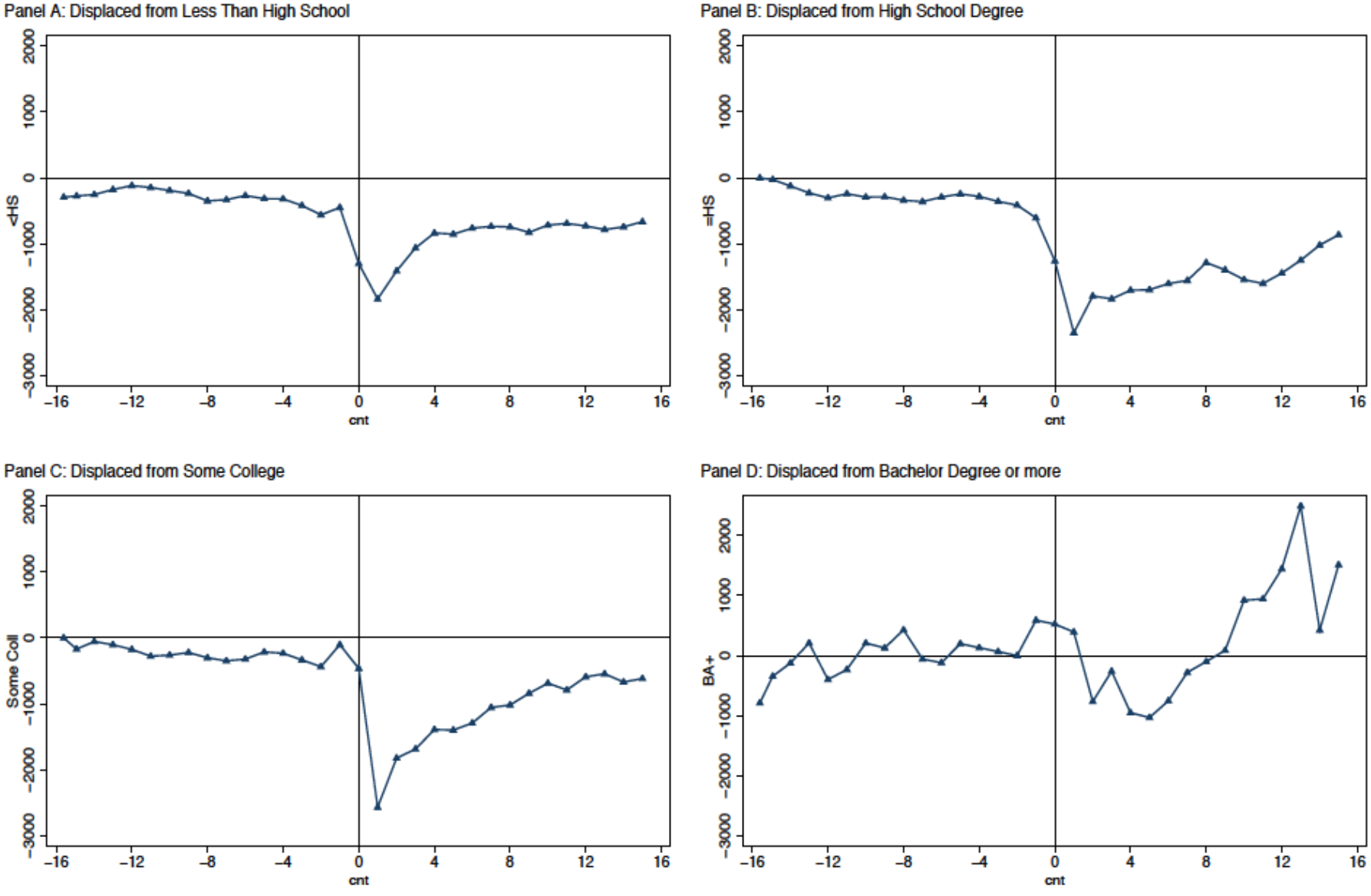
Other Studies of Job Displacement

- Several recent papers by von Wachter and co-authors use JLS-style methods to look at job displacement in other contexts
  - von Wachter, Song and Manchester (2009): Use social security records to study long-term effects of displacement in 1982 recession (more states, 20 years of follow-up)
  - von Wachter and Handwerker (2009): Use California UI records matched to CPS interviews. Allows examination of heterogeneity by education
  - Sullivan and von Wachter (2009): Match JLS Pennsylvania data to Social Security death records to look at mortality
  - von Wachter, Bender and Schmeider (2009): Use German social security data to look at 1982 recession in Germany. Includes detailed information (e.g. days worked) on each employment spell, so allows an analysis of wages

Figure 3: Earnings Losses at Job Separation 1980–1986 vs. Non–Separators
Earnings at All Jobs Including Zeros, Men in Stable Job 1974–1979 (in $1000)



Source: 1% Files of Social Security administrative data (see text). Earnings in 2000 Dollars.



Source: Merged California UI–BW and March CPS Data (see text).

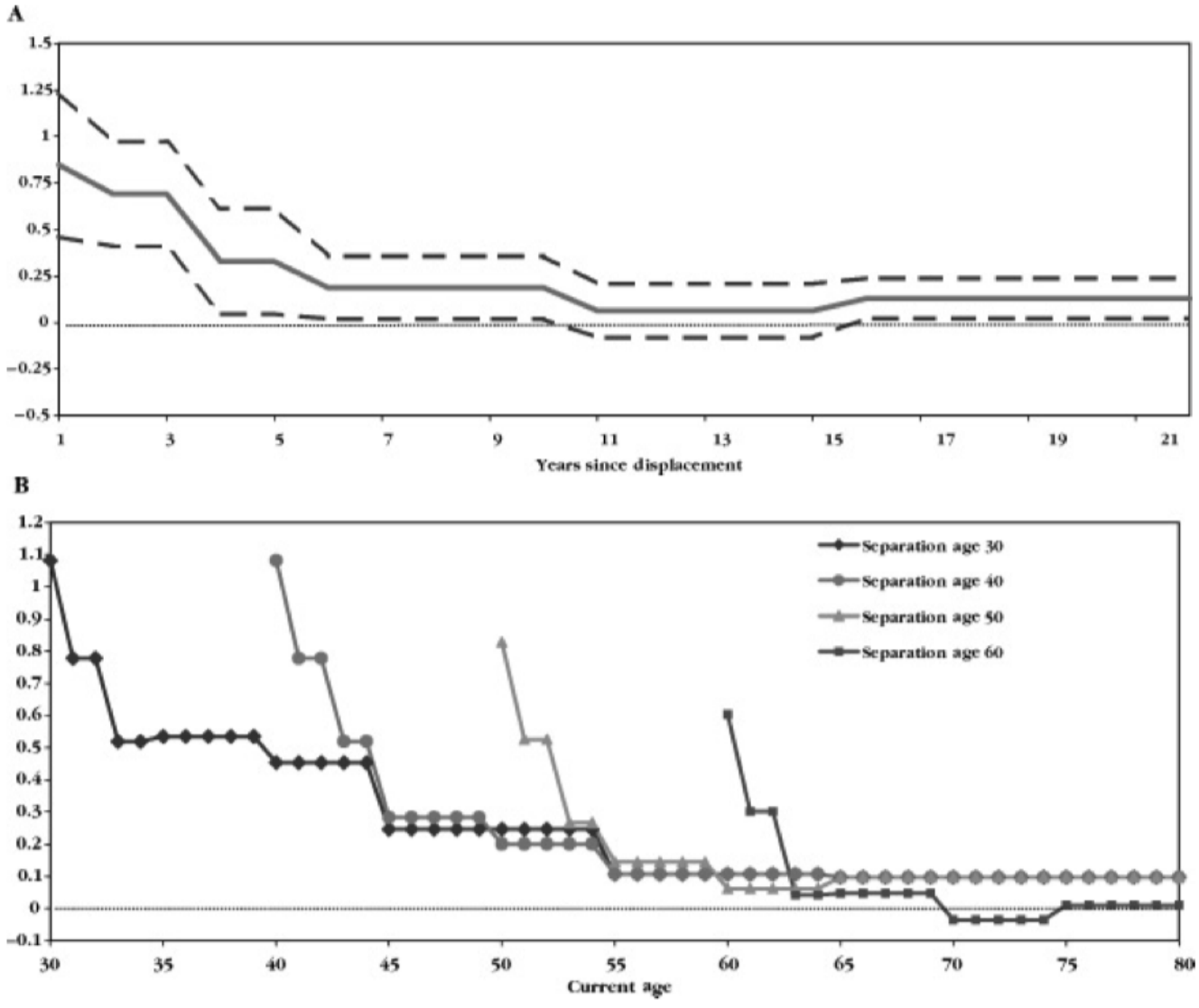


FIGURE II

The Effect of Displacement on Log-Odds of Death by Years since Displacement (Sample of Men in Stable Employment 1974–1979, Firm 1979 Employment ≥50, No Further Presence Restriction in PA Labor Market)

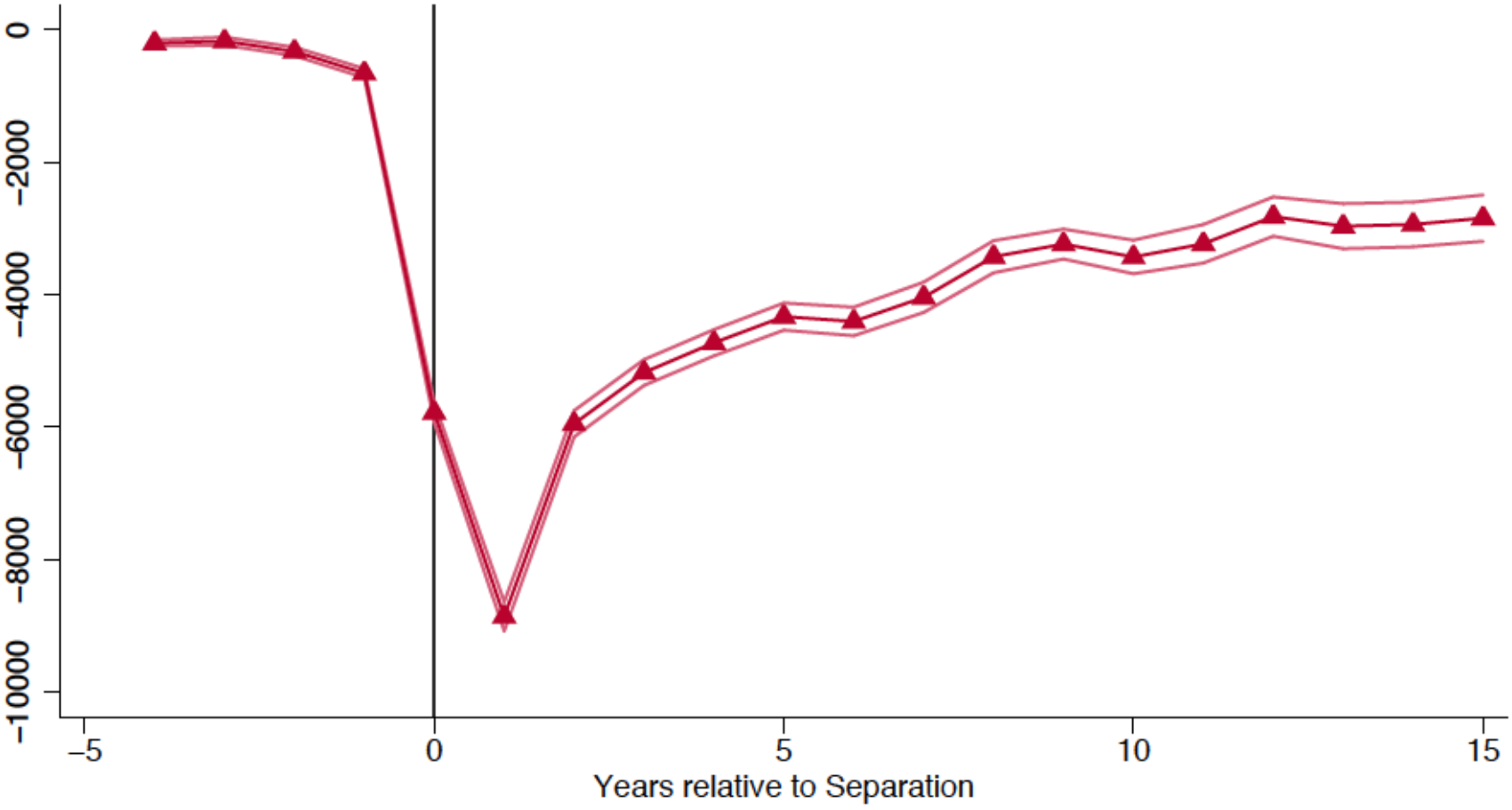


Figure 5: Total Yearly Earnings of Displaced Workers relative to Non−Displaced

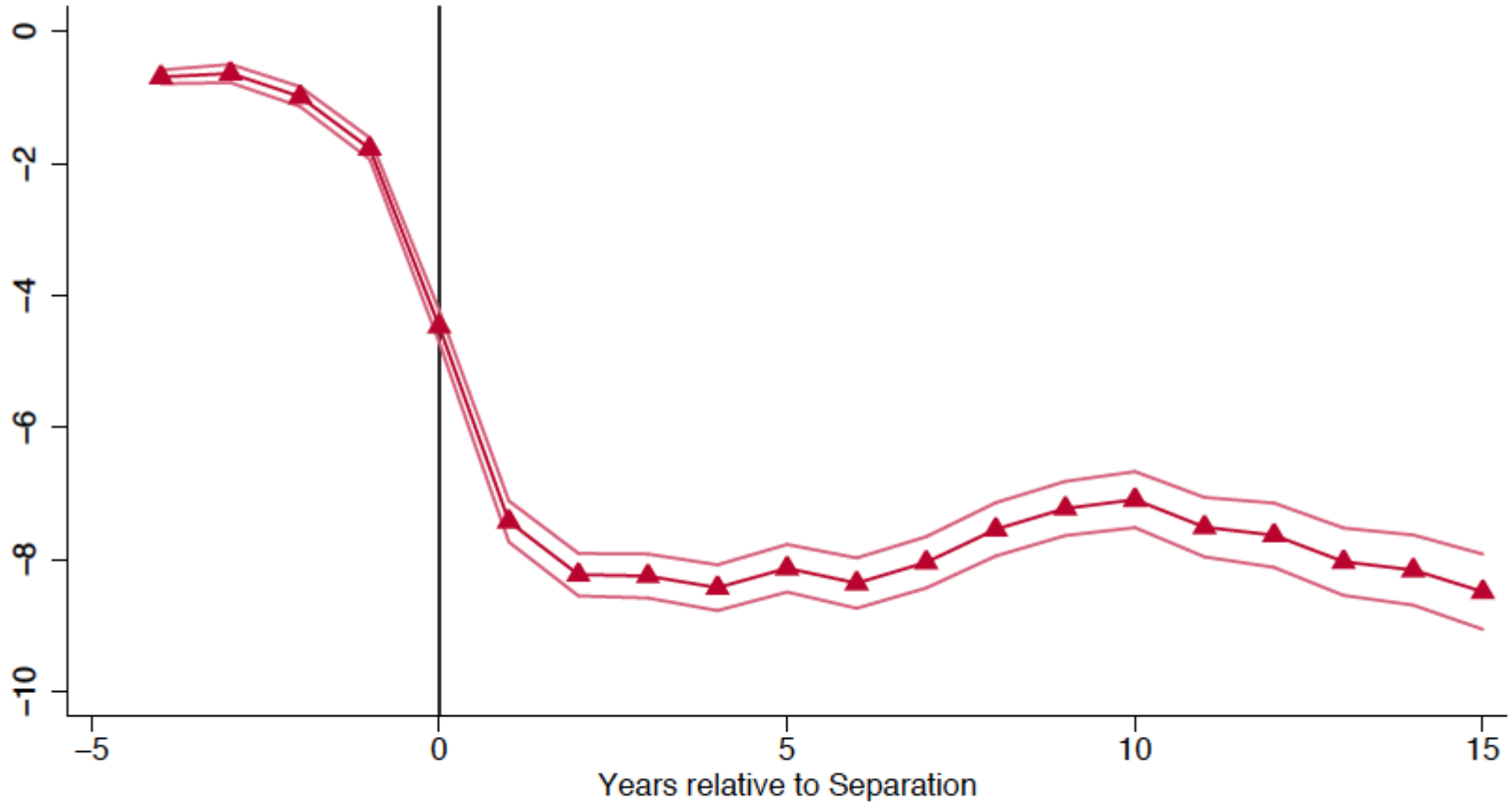
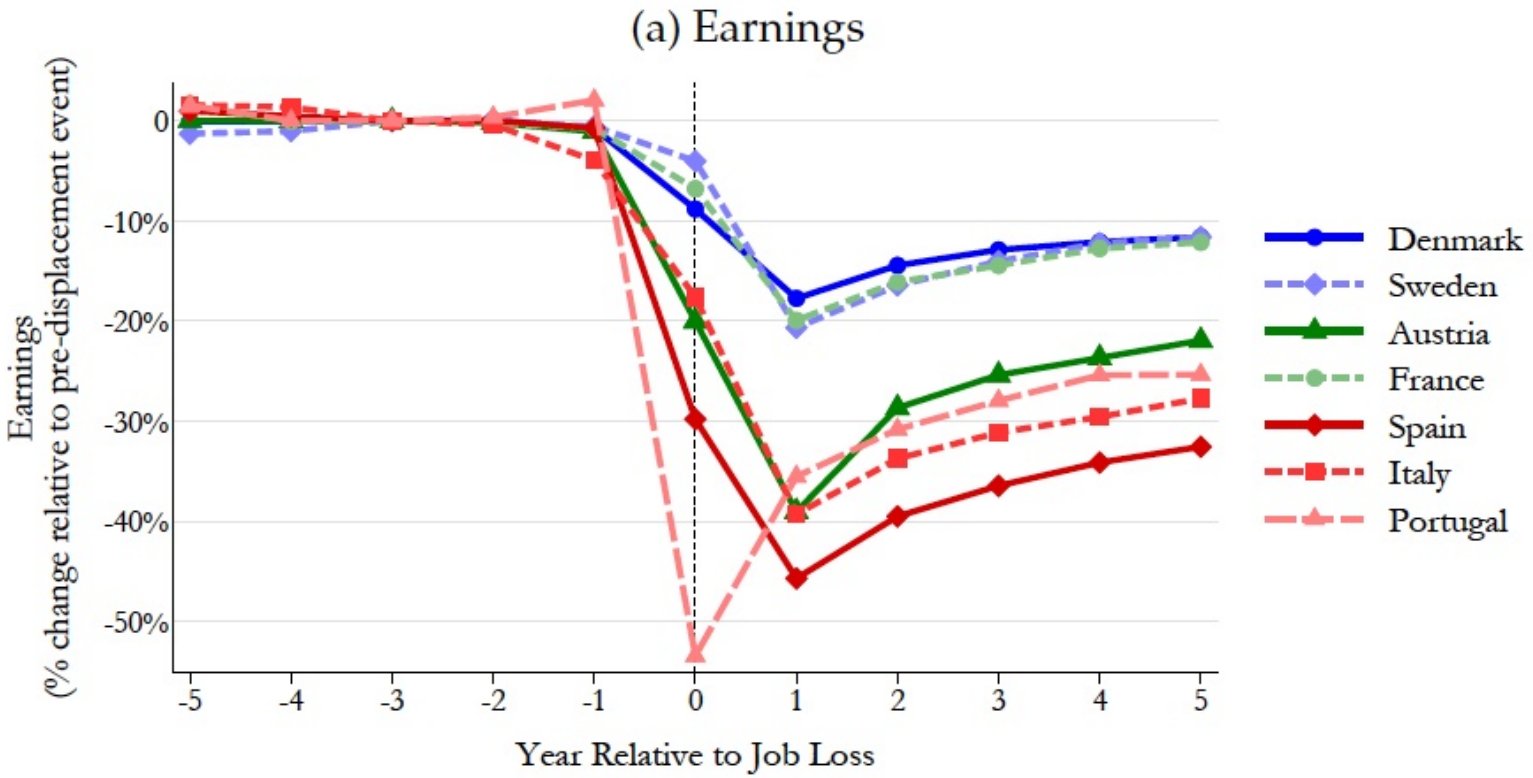
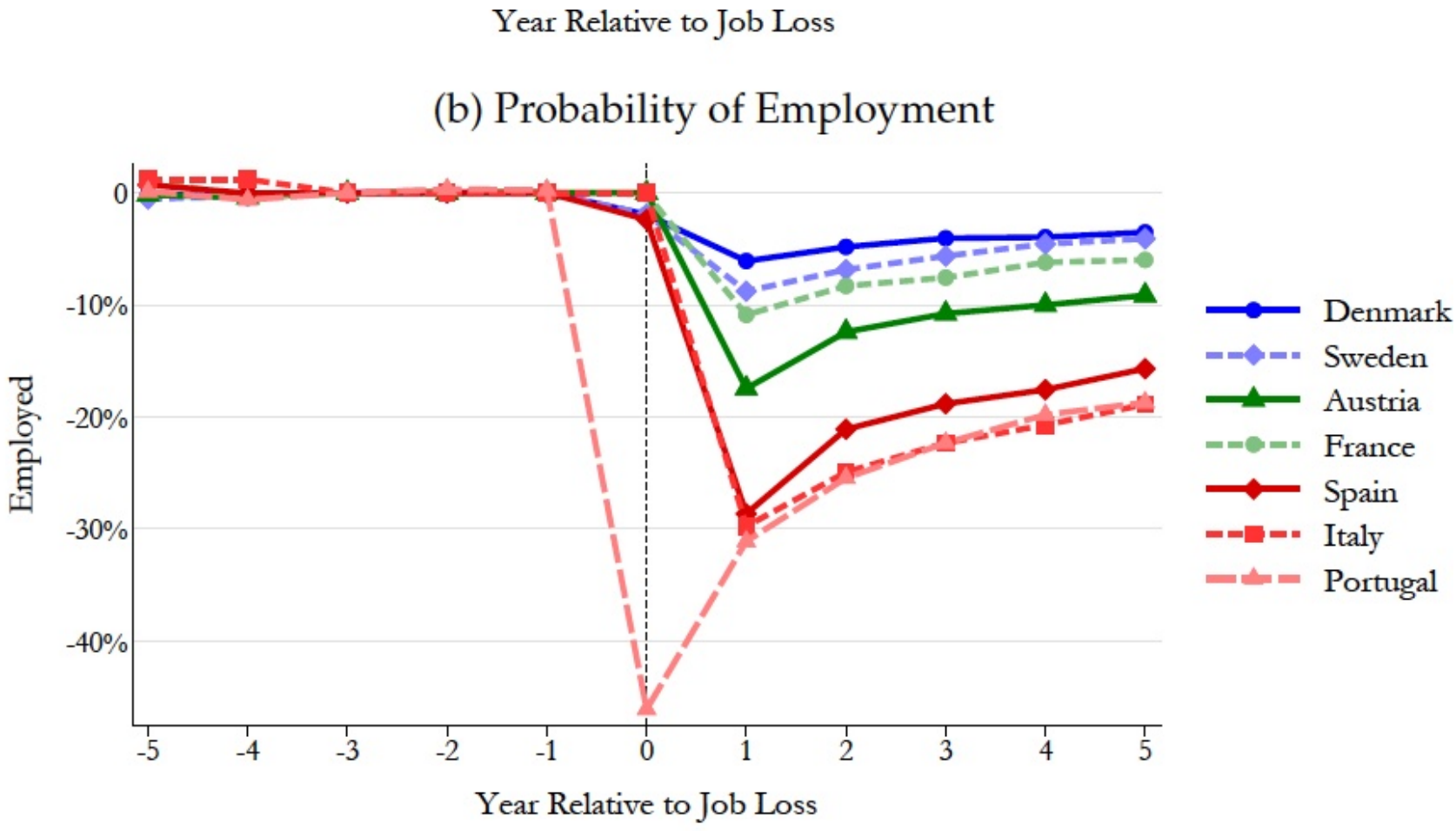


Figure 6: Daily Wage of Displaced Workers relative to Non−Displaced

Evidence from Europe: Bertheau et al. (2023)



Evidence from Europe: Bertheau et al. (2023)



Discussion: Why these enormous effects after losing a job?

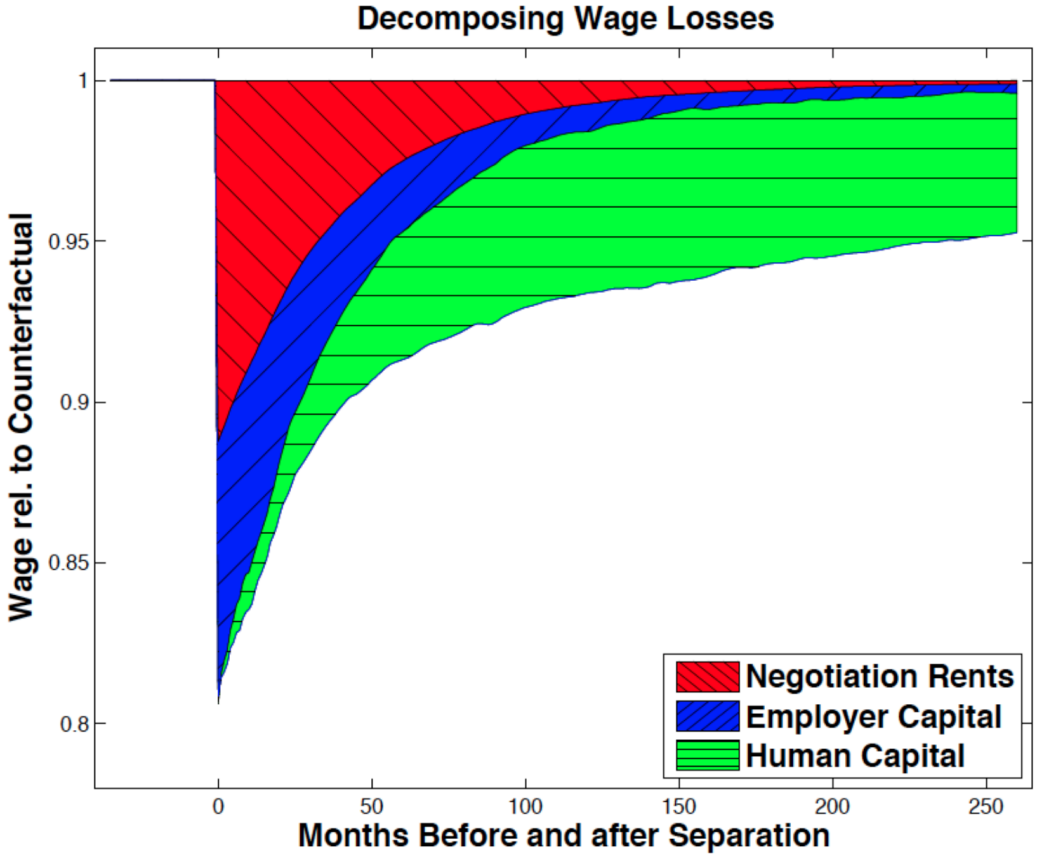
- Ideas?

Discussion: Why these enormous effects after losing a job?

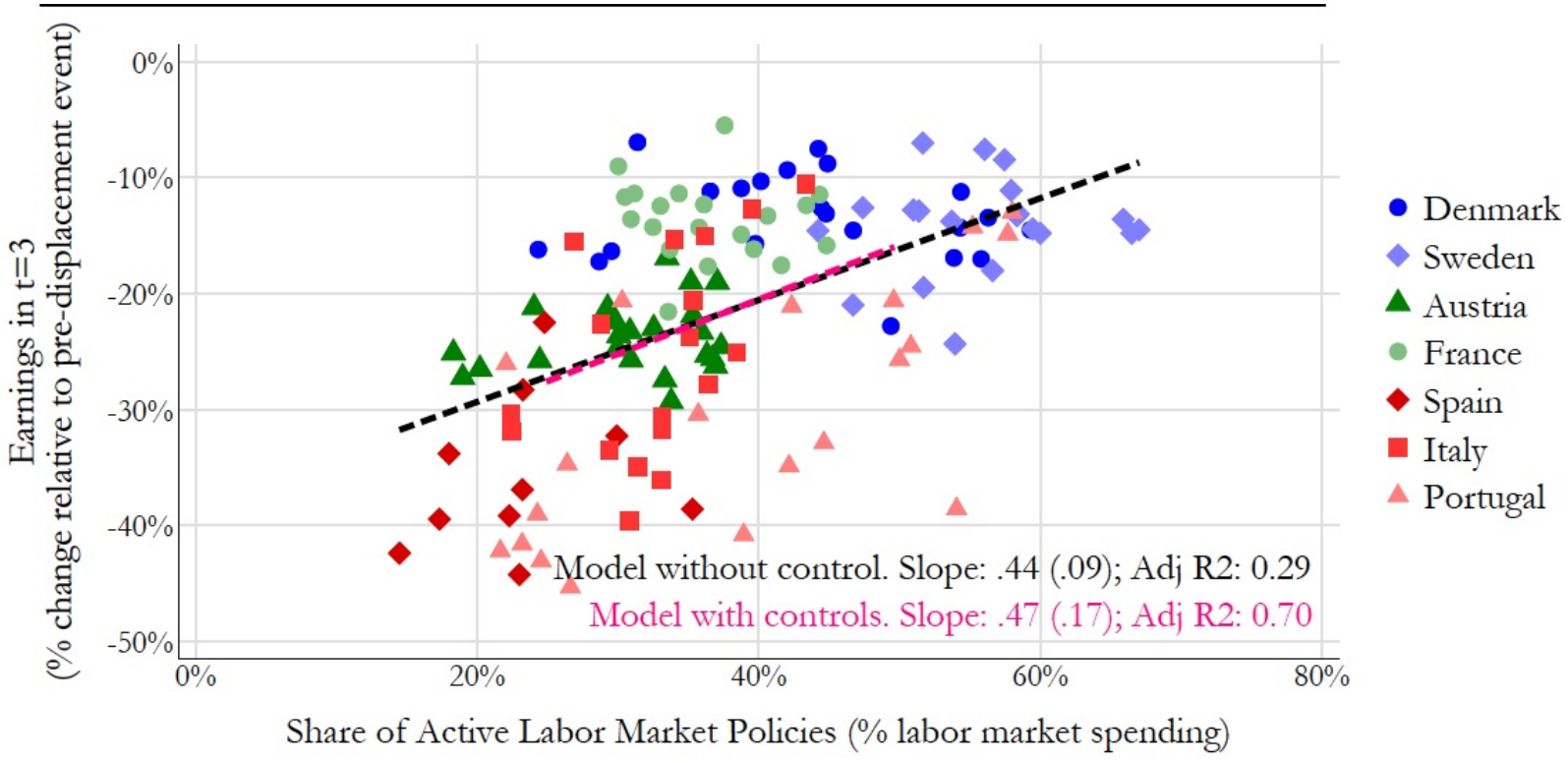
- Ideas?
- Jarosch JMP is a nice attempt: structural model that allows for heterogenous jobs in both security and wages.
- Unemployed workers fell off the ladder: sorted into worst jobs + possible depreciation of skills + loss of bargaining chip.

Discussion: Why these enormous effects after losing a job?

- Ideas?
- Jarosch JMP is a nice attempt: structural model that allows for heterogenous jobs in both security and wages.
- Unemployed workers fell off the ladder: sorted into worst jobs + possible depreciation of skills + loss of bargaining chip.



But labor market institutions are also important!



Notes: In the figure, the y-axis represents the estimated displacement effect on labor earnings 3 years after displacement (the coefficient θ_3 when estimating Equation 1 separately by country and year of mass layoff) and on the x-axis we have the percentage of spending on ALMPs out of the total labor programs spending for a give country-year-of-mass-layoff. Linear fit lines show the slope regression coefficient (black line) or the residualized regression coefficient (pink line) after controlling for country fixed effects, workers and employers' controls, macroeconomic controls, country fixed effects, and additional labor market institutions. The full set of labor market institutional controls is: difference in OECD harmonized score of permanent versus temporary workers protection, share of involuntary part-time workers, share of temporary employment, UI duration, UI replacement rate, UI strictness of search requirements, union density and coverage, and total expenditures in labor market policies as share of GDP. The last column reports post-lasso estimates, i.e. applies OLS to the variables selected by the first-stage variable selection method (Lasso) applied to the full set of institutional variables described above. Standard errors are clustered at the country level.

Discussion

- Displacement costs are real.
- Need to re-think carefully about impact of labor demand shocks and/or policies.
- What are the labor market consequences to interval/external shocks that might destroy jobs?
- China Shock from the presentation that we have seen in class a few weeks ago is one example.

- Another useful example: Green Policies.

Walker (2013)

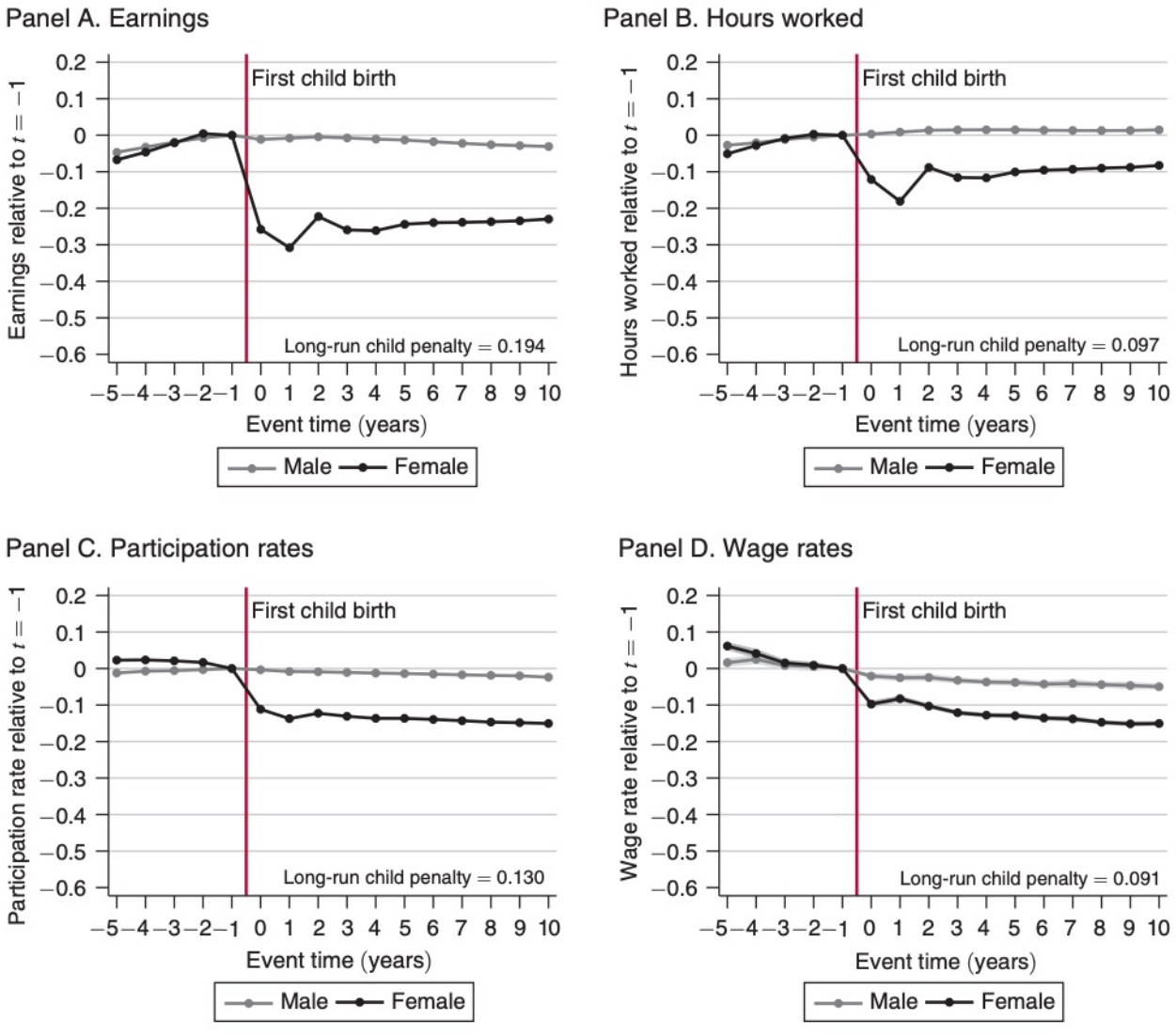
THE TRANSITIONAL COSTS OF SECTORAL REALLOCATION: EVIDENCE FROM THE CLEAN AIR ACT AND THE WORKFORCE*

W. REED WALKER

This article uses linked worker-firm data in the United States to estimate the transitional costs associated with reallocating workers from newly regulated industries to other sectors of the economy in the context of new environmental regulations. The focus on workers rather than industries as the unit of analysis allows me to examine previously unobserved economic outcomes such as nonemployment and long-run earnings losses from job transitions, both of which are critical to understanding the reallocative costs associated with these policies. Using plant-level panel variation induced by the 1990 Clean Air Act Amendments (CAAA), I find that the reallocative costs of environmental policy are significant. Workers in newly regulated plants experienced, in aggregate, more than $5.4 billion in forgone earnings for the years after the change in policy. Most of these costs are driven by nonemployment and lower earnings in future employment, highlighting the importance of longitudinal data for characterizing the costs and consequences of labor market adjustment. Relative to the estimated benefits of the 1990 CAAA, these one-time transitional costs are small. *JEL* Codes: Q50, H41, R11.

Effects of Child-Birth: Kleven-Landais-Sogaard, 2019

Effects of Child-Birth: Kleven-Landais-Sogaard, 2019



No sign of convergence 20 years after

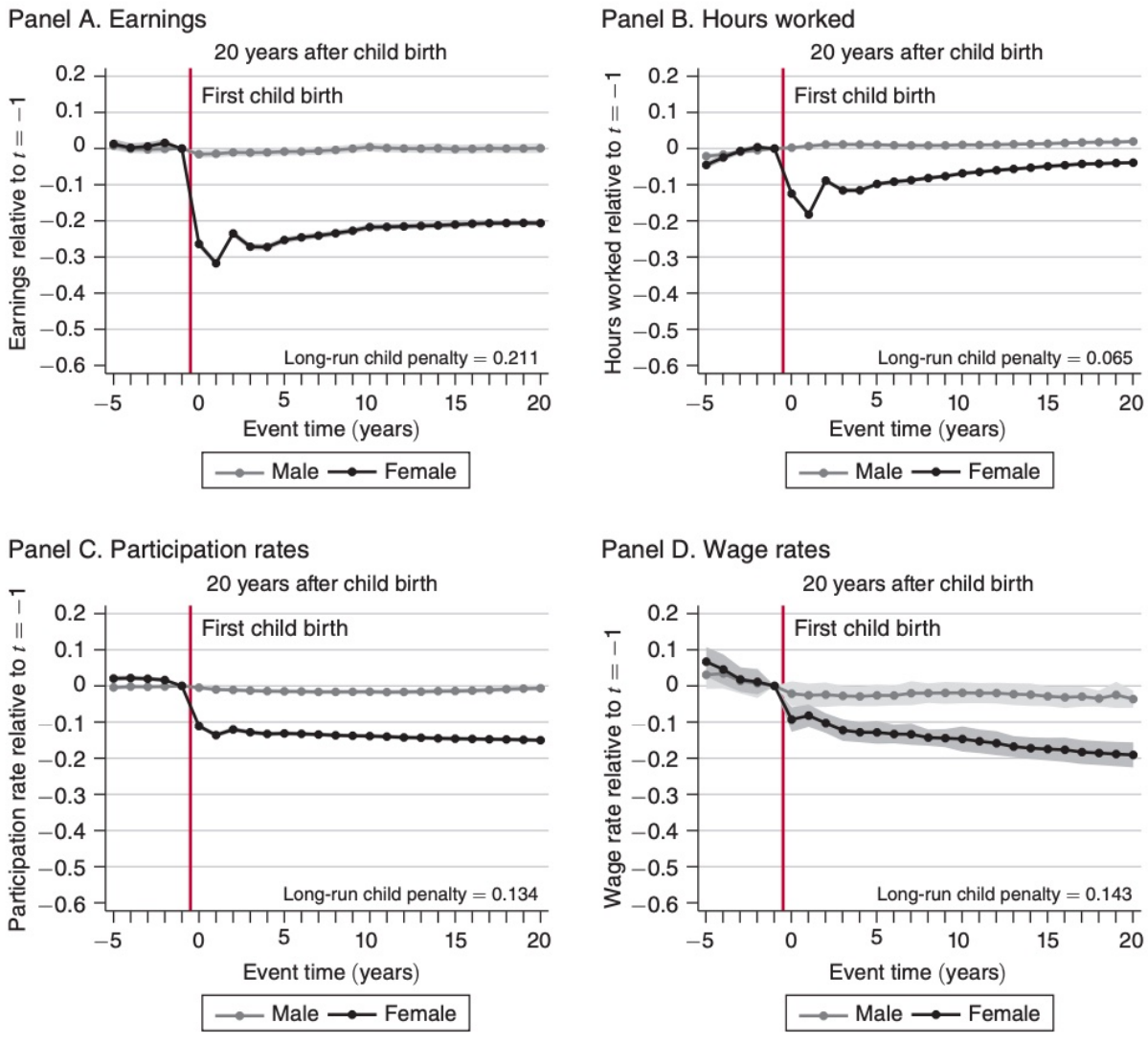


FIGURE 2. IMPACTS OF CHILDREN IN THE VERY LONG RUN

Culture Matters?

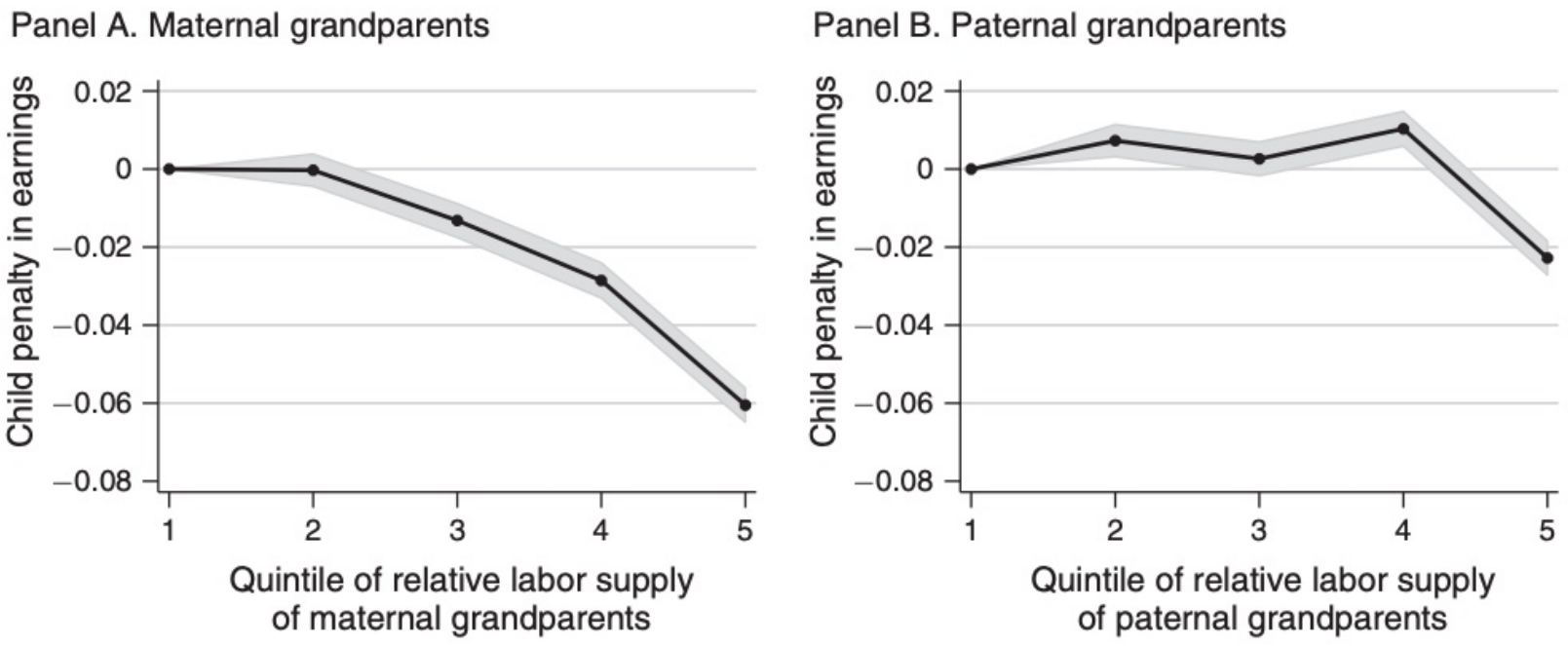
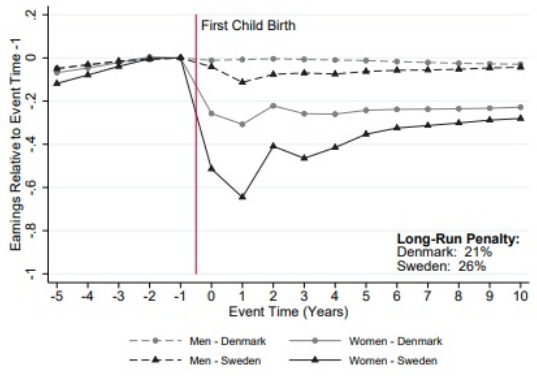


FIGURE 6. INTERGENERATIONAL TRANSMISSION OF CHILD PENALTIES WITHOUT CONTROLS

Notes: The figure shows the child penalty in earnings against quintiles of the relative labor supply distribution of the grandparents. The relative labor supply of grandparents is based on cumulated hours worked over the period 1964–1979 (obtained from ATP pension contributions). The child penalties are estimated using equation (5) and the statistic reported is P_q, as defined in Section VB. The panels only include controls for the birth cohort of the grandparents. The shaded 95 percent confidence intervals are based on robust standard errors.

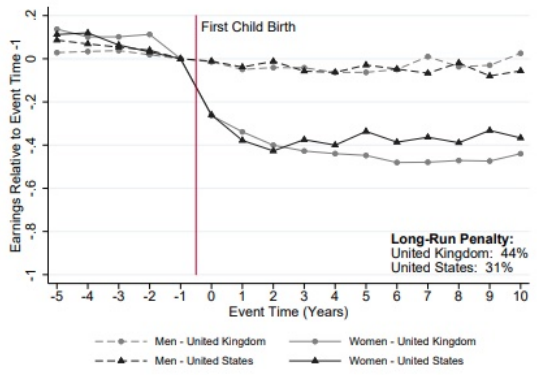
Evidence from various countries

Figure 1: Child Penalties in Earnings in Scandinavian Countries



Notes: The figure shows percentage effects of parenthood on earnings across event time t for each gender g, i.e. P_t^g defined above. The figure also displays long-run child penalties, defined as the average penalty P_t from event time 5 to 10. Earnings are unconditional on employment status and the effects therefore include both the extensive and intensive margins.

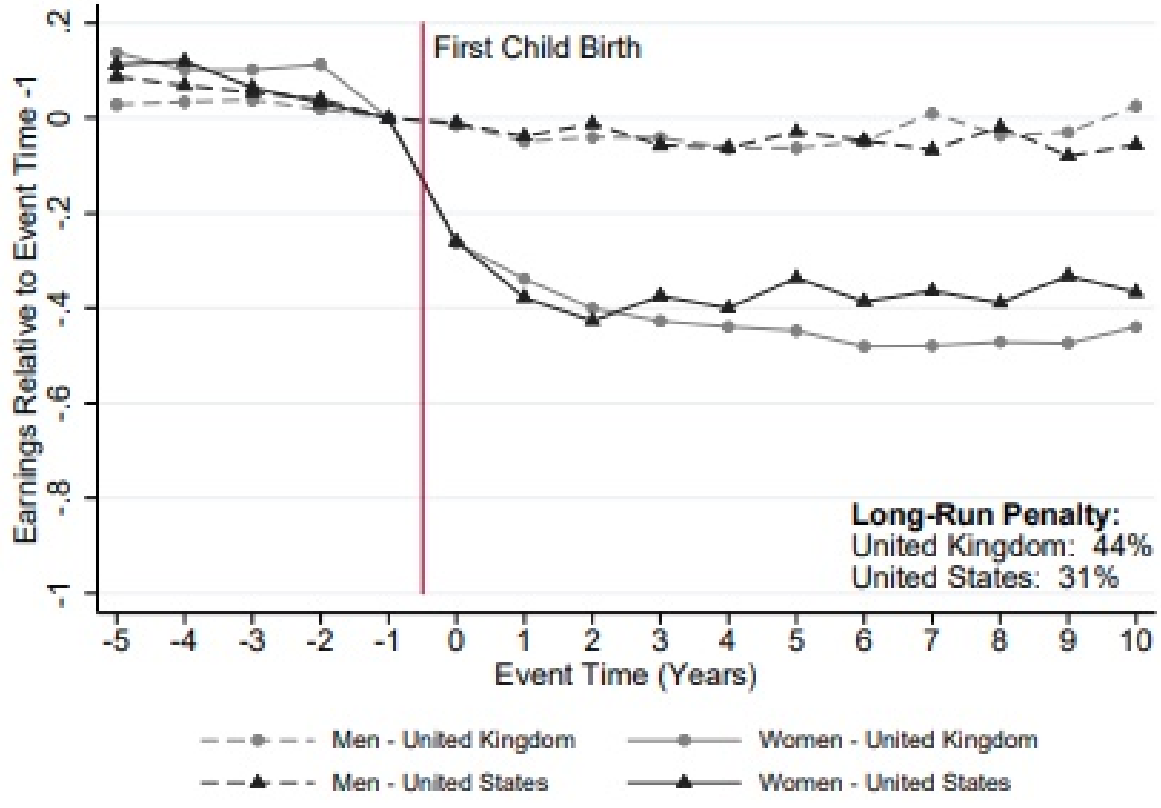
Figure 2: Child Penalties in Earnings in English-Speaking Countries



Notes: See the notes to Figure 1.

Evidence from various countries

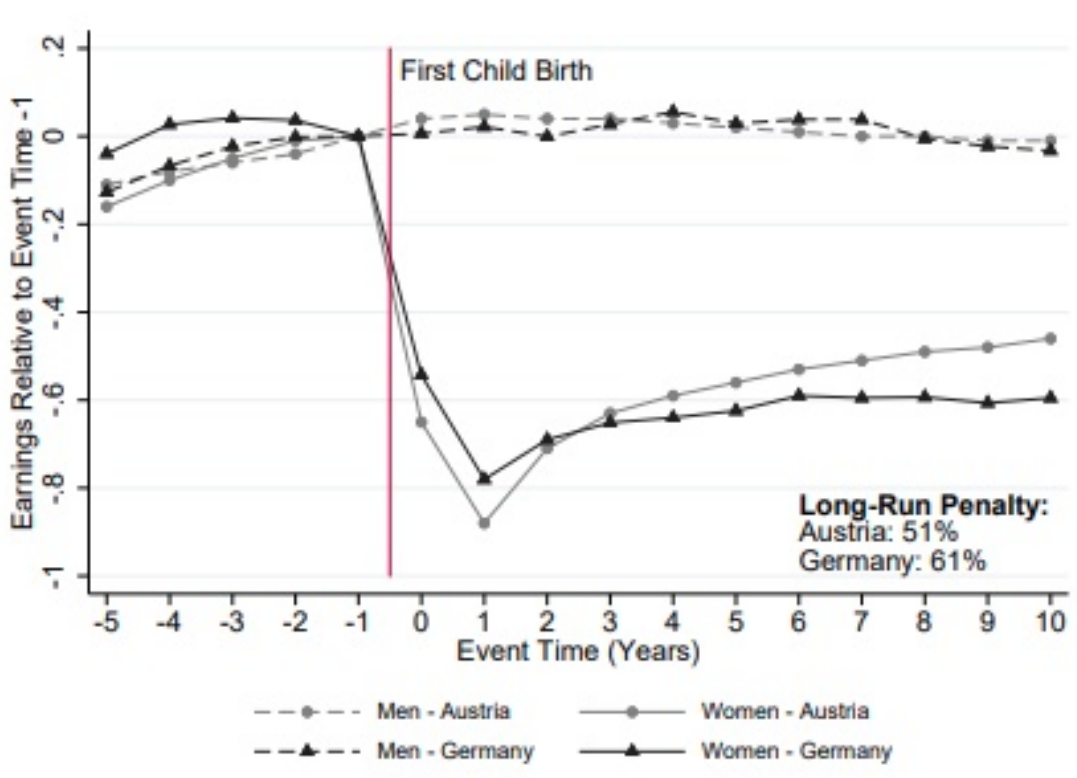
Figure 2: Child Penalties in Earnings in English-Speaking Countries



Notes: See the notes to Figure 1.

Evidence from various countries

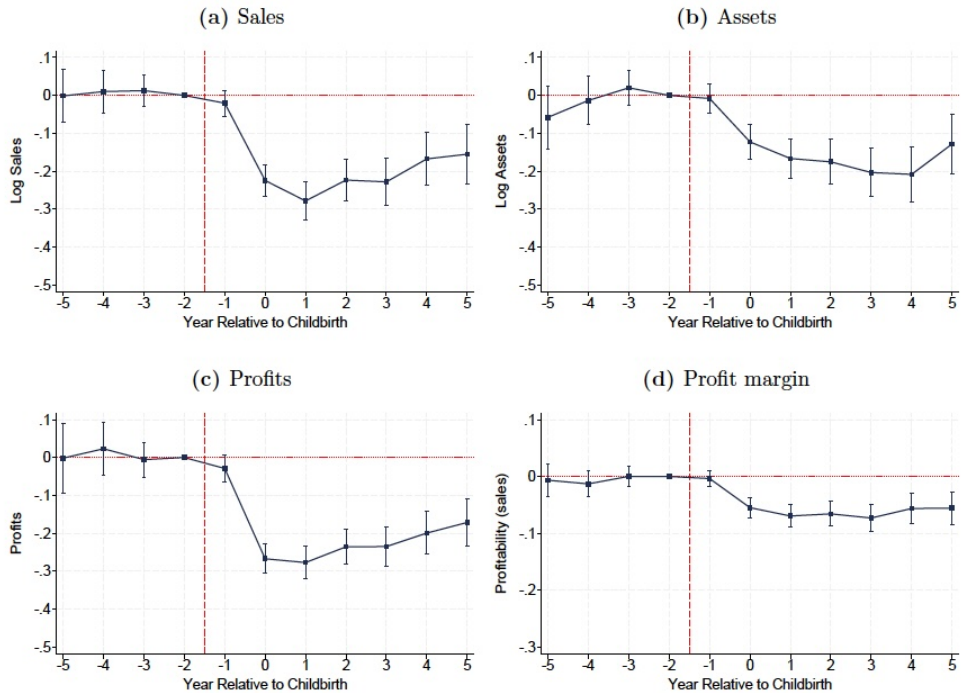
Figure 3: Child Penalties in Earnings in German-Speaking Countries



Notes: See the notes to Figure 1.

Childbirth of Women Entrepreneurs spillovers into the performance of the firm

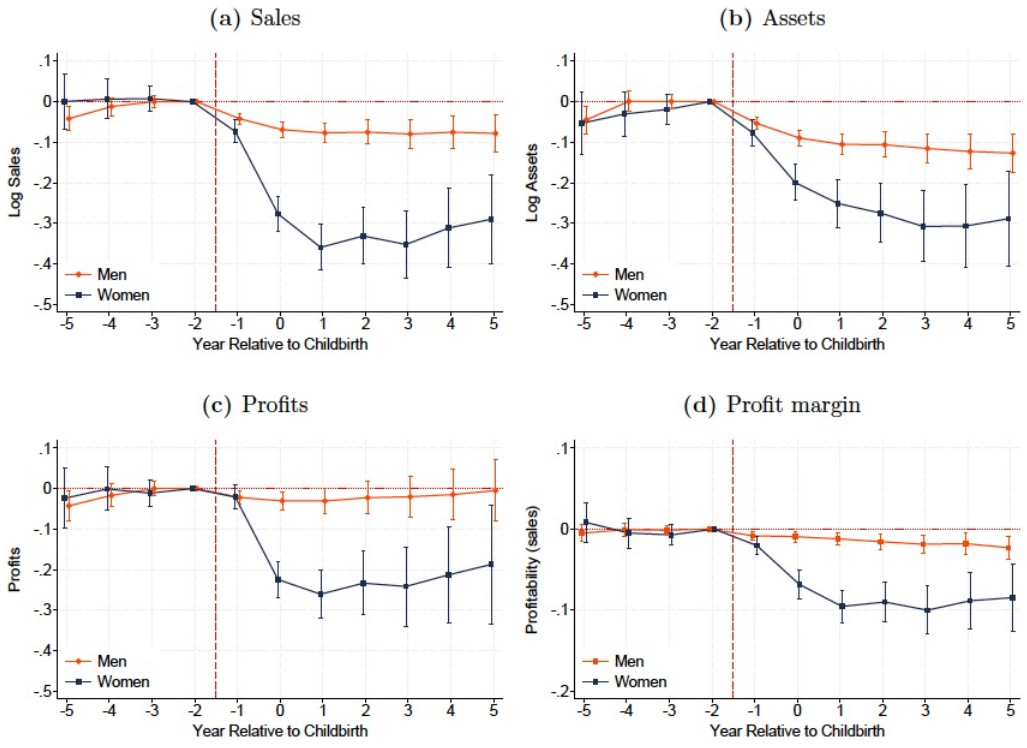
Figure 4: Firm outcomes: matched sample



Notes: The graphs show event-study estimates obtained by fitting equation 2 on log sales (panel (a)), log assets (panel (b)), profits (panel (c)), and profit margin (panel (d)). Year 0 is when the first childbirth event takes place. The treated group is a sample of firms owned by a female entrepreneur for at least two years before she had her first child. The control group is a matched sample of firms owned by female entrepreneurs with zero observed fertility. Coefficients for profits are reported as a percentage of the counterfactual outcome absent children. Profit margin is the ratio of profits to sales. Control variables include indicators for firm age, the number of firm owners, a polynomial for individual age, and marital status. Firm effects and industry × province × year fixed effects are included. I report 95% confidence intervals based on standard errors clustered at the firm level.

Much more muted effects for men!

Figure 6: Firm outcomes: mothers vs. fathers



Notes: The graphs show event-study estimates obtained by fitting equation 5 on log sales (panel (a)), log assets (panel (b)), profits (panel (c)), and profit margin (panel (d)), separately for mothers and fathers. Year 0 is when the first childbirth event takes place. Coefficients for profits are reported as a percentage of the counterfactual outcome absent children. Profit margin is the ratio of profits to sales. Control variables include indicators for firm age, the number of firm owners, a polynomial for individual age, and marital status. Firm effects and industry × province × year fixed effects are included. I report 95% confidence intervals based on standard errors clustered at the firm level.

Appendix

A simple Search Model

- Time is discrete.
- Unemployed individual who is infinitely lived, risk neutral, with discount factor: $\beta = \frac{1}{1+r}$
- Each period receives a job offer that she can accept/reject.
- If job is accepted, it lasts forever. There is no on the job search.
- Individual employed at wage w as flow utility w.
- Unemployed individual receives benefit b and has flow utility ν.
- Wages are drawn from distribution $F(w)$ with pdf $f(w)$.

Value Function / Bellman Equation

- We can set up the bellman equation that defines the value of unemployment

$$V = b - \nu + \beta \int_0^\infty max(V, \frac{w}{1-\beta}) f(w) dw \tag{9}$$

- This expression can be written as

$$\begin{aligned} V &= b - \nu + \beta \int_0^\infty \left(V + max(0, \frac{w}{1-\beta} - V) \right) f(w) dw \\ V(1-\beta) &= b - \nu + \frac{\beta}{1-\beta} \int_0^\infty \left(max(0, w - (1-\beta)V) \right) f(w) dw \end{aligned} \tag{10}$$

- What is the reservation wage? $\frac{w^*}{1-\beta} = V$
- Plugging this expression we then get:

$$w^* = b - \nu + \frac{1}{r} \int_0^\infty (w - w^*) f(w) dw \tag{11}$$

- which can be restated as:

$$w^* - (b - \nu) = Pr(w \geq w^*) \times \frac{\mathrm{E}[w - w^* | w > w^*]}{r} \tag{12}$$

which coincides with equation (**??**)

Jarosch (2015)

- Jarosch (2015) approaches job displacement with a fresh theoretical eye
- The long-term "scarring" effect of job displacement is a major reason we're concerned with unemployment
- But most modern workhorse models used to study unemployment (e.g. DMP search models) do not generate an unemployment scar
- Jarosch proposes a search model that can generate long-term effects of job displacement
- Then applies the model to the German social security data studied by von Wachter and co.

Jarosch (2015): Model

- Jarosch proposes a "job ladder" model with on-the-job search
- Workers search for new jobs while employed or unemployed
- Jobs are heterogeneous on two dimensions:
  - Productivity
  - Job security (expected length of match)
- Workers "climb" the ladder, searching for better and better jobs – implies unemployment begets future unemployment spells
- Model also allows employment history to alter workers' bargaining position, and general human capital to evolve differentially for employed and unemployed
- Three mechanisms here that can generate persistent displacement effects: position in the job ladder, experience, bargaining position

Setup

- Firm types: $\theta = [\theta_y, \theta_\delta]$
  - θ_y is productivity
  - θ_δ is job distruction probability (1 - job security)
- Workers are homogeneous ex ante
- Unemployed workers meet firms with probability λ_0, employed workers λ_1

Wages and Value Functions

- Fixed wage contracts, re-bargained when either party has a credible threat (sequential auction framework of Cahuc, Postel-Vinay and Robin, 2006)
- $w\left(\theta, \hat{\theta}\right)$: Worker's wage when matched with type θ, and $\hat{\theta}$ used as outside option in last negotiation (negotiation benchmark)
- Notation for value functions:
  - $W(\theta, \hat{\theta})$: employed worker's value
  - U: unemployed worker's value
  - J: value of a job
  - $S(\theta)$: joint surplus of a match between a worker and firm θ

Wages

- For bargaining parameter $\alpha \in [0, 1]$, wage for a newly employed worker at firm type θ will be negotiated to satisfy

$$W(\theta, u) - U = \alpha S(\theta)$$

- Suppose a worker at firm type θ_1 is offered a job at outside firm type θ_2
- If $S(\theta_2) > S(\theta_1)$, the worker transfers to θ_2 and θ_1 becomes the negotiation benchmark
- New wage $w(\theta_2, \theta_1)$ is negotiated to satisfy

$$W(\theta_2, \theta_1) - U = S(\theta_1) + \alpha \left(S(\theta_2) - S(\theta_1) \right)$$

- Worker gains a fraction α of the extra surplus generated by the better match

Wages

- If $S(\theta_2) < S(\theta_1)$, the worker will not transfer, but if θ_2 is better than previous negotiation benchmark may renegotiate wage at θ_1 so that

$$W(\theta_1, \theta_2) - U = S(\theta_2) + \alpha \left(S(\theta_1) - S(\theta_2) \right)$$

- This suggests the following classification of firms:
  - $M_1(\theta)$: Firm types x such that $S(x) > S(\theta)$
  - $M_2(\theta, \hat{\theta})$: Firm types x such that $S(\theta) > S(x) > S(\hat{\theta})$
- A worker at θ with negotiation benchmark $\hat{\theta}$ will move if offered something in $M_1(\theta)$, stay but renegotiate if offered something in $M_2(\theta, \hat{\theta})$, and ignore if offered anything else

Negotiation Capital

- In the sequential auction framework, workers build "negotiation capital" through repeated sampling of outside offers
- Formally, negotiation capital is $W(\theta, \hat{\theta}) - W(\theta, u)$
- This is the extra value a worker gets from θ because s/he is negotiating based on the benchmark $\hat{\theta}$ rather than unemployment
- This can generate large cross-sectional wage dispersion
- Also has the potential to generate sharp wage reductions following an unemployment spell, as negotiation capital is destroyed when a worker enters unemployment

Value Function for an Employed Worker

$$W(\theta,\hat{\theta}) = w(\theta,\hat{\theta}) + \beta \times \{\ (1-\theta_\delta)$$

$$\times[\ \lambda_1 \left(\int_{x \in M_1(\theta)} W(x,\theta) dF(x) + \int_{x \in M_2(\theta,\hat{\theta})} W(\theta,x) dF(x) \right)$$

$$+ \left(1 - \lambda_1 \int_{x \in M_1(\theta) \cup M_2(\theta,\hat{\theta})} dF(x) \right) W(\theta,\hat{\theta})\ ] + \theta_\delta U\ \}.$$

- Flow utility $w(\theta,\hat{\theta})$, discount rate β
- With probability θ_δ, the match ends and the worker enters unemployment
- With probability $(1-\theta_\delta)$, the match doesn't end, and
  - With probability λ_1, the worker meets a new firm
    - In $M_1(\theta)$: Switch and get $W(x,\theta)$
    - In $M_2(\theta,\hat{\theta})$: Don't switch but renegotiate for $W(\theta,x)$
    - In neither: Stick with $W(\theta,\hat{\theta})$
  - With probability $1-\lambda_1$, the worker does not meet a firm and keeps $W(\theta,\hat{\theta})$

Value Function for an Unemployed Worker

$$U = z + \beta \left[\lambda_0 \int_{x \in M_1(u)} W(x,u) dF(x) + \left(1 - \lambda_0 \int_{x \in M_1(u)} dF(x) \right) U \right]$$

- Flow utility is unemployment benefit z
- With probability λ_0, the worker meets a new firm
  - In $M_1(u)$: Take the job and get $W(x,u)$
  - Not in $M_1(u)$: Reject job, keep U
- With probability $1 - \lambda_0$, the worker doesn't meet a new firm, and keeps U

Value Function for a Firm

$$J(\theta,\hat{\theta}) = \theta_y - w(\theta,\hat{\theta}) + \beta(1-\theta_\delta) \times [\ \lambda_1 \int_{x \in M_2(\theta,\hat{\theta})} J(\theta,x) dF(x)$$

$$+ \left(1 - \lambda_1 \int_{x \in < M_1(\theta) \cup M_2(\theta,\hat{\theta})} dF(x)\right) J(\theta,\hat{\theta})\ ]$$

- Flow utility for a firm of type θ employing a worker with benchmark $\hat{\theta}$ is productivity θ_y minus wage $w(\theta,\hat{\theta})$
- With probability θ_δ, job is destroyed and has no continuation value
- With probability $(1-\theta_\delta)$, job continues, and
  - With probability λ_1, worker finds a new job
    - In $M_1(\theta)$: Job destroyed, no continuation value
    - In $M_2(\theta,\hat{\theta})$: Renegotiate, get $J(\theta,x)$
    - In neither: Stick with $J(\theta,\hat{\theta})$
  - With probability $1-\lambda_1$, worker does not find a job and firm keeps $J(\theta,\hat{\theta})$

Match Surplus

- Match surplus for the worker/firm pair is

$$S(\theta, \hat{\theta}) = \max\left\{0, W(\theta, \hat{\theta}) - U + J(\theta, \hat{\theta})\right\}$$

- Plugging in value functions yields

$$S(\theta) = \theta_y - z + \beta \times \{ \ (1 - \theta_\delta) \left(S(\theta) + \alpha\lambda_1 \int_{x \in M_1(\theta)} (S(x) - S(\theta)) \, dF(x) \right)$$

$$-\alpha\lambda_0 \int_{x \in M_1(u)} S(x) dF(x) \}$$

- Result: Surplus doesn't depend on $\hat{\theta}$, so $S(\theta, \hat{\theta}) = S(\theta)$
- Makes sense – negotiation benchmark just determines split of the surplus
- This equation can be solved numerically, given parameter values

Comparative Statics

- Model gives rise to some intuitive comparative statics
  - Wages increasing in θ_δ (compensating differential)
  - θ_y has an ambiguous effect on wages, depending on α
    - When $\alpha = 0$, moving workers get exactly their old match's surplus. Higher productivity firms offer higher growth potential and therefore lower wages
    - When $\alpha = 1$, workers capture the entire surplus and higher productivity firms pay higher wages
- If $E[\theta_\delta|\theta_y]$ and $E[\theta_y|\theta_\delta]$ are strictly increasing:
  - Job security and productivity are increasing in employment tenure and job tenure
  - Hazard to unemployment is decreasing in employment tenure and job tenure

Efficiency

- Jarosch assumes $\lambda_1 < \lambda_0$. This implies the model's equilibrium is inefficient if $\alpha < 1$

Efficiency

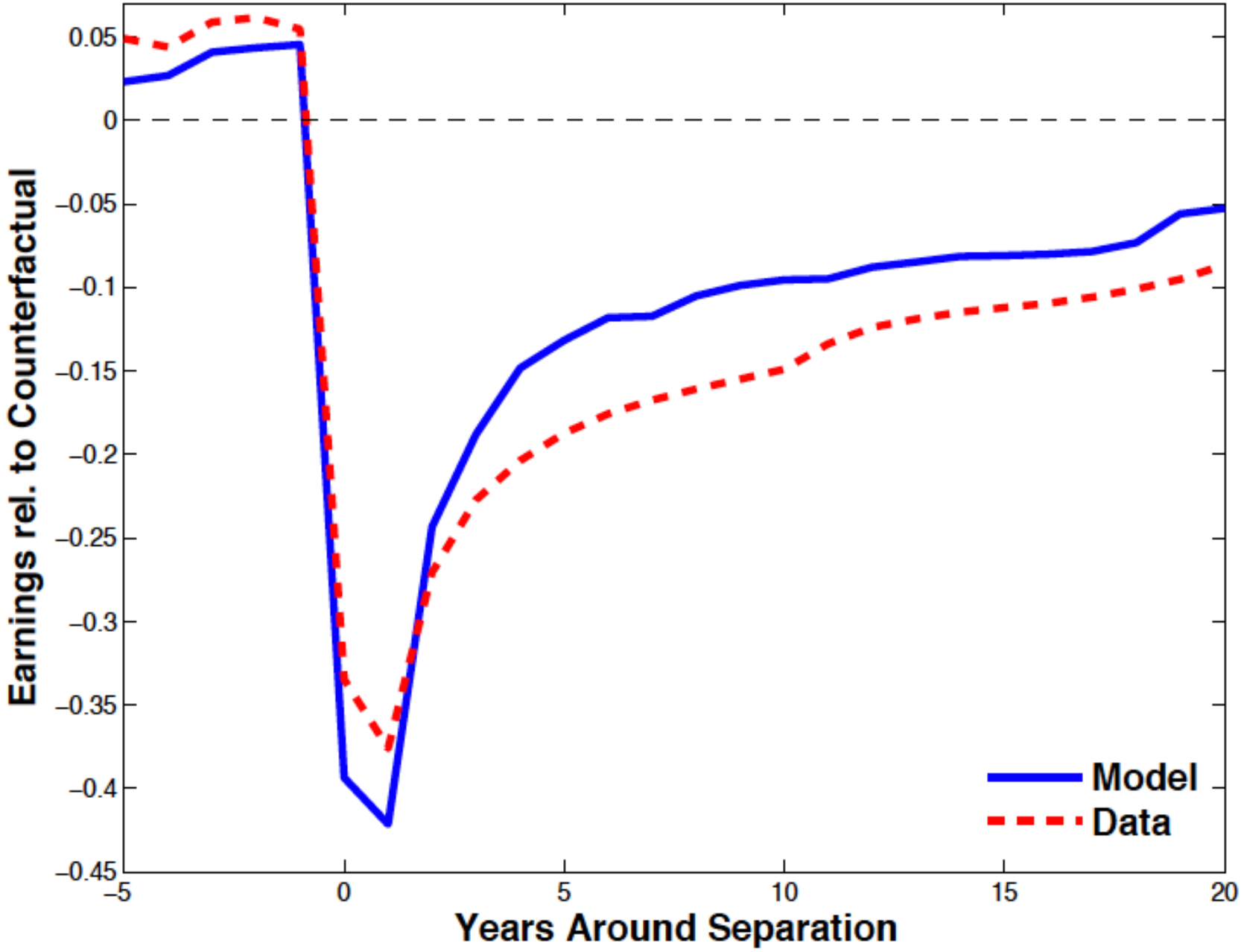
- Jarosch assumes $\lambda_1 < \lambda_0$. This implies the model's equilibrium is inefficient if $\alpha < 1$
- Social planner places more value on unemployment than does a matched worker/firm pair, because unemployed workers are available to match with other firms, which is not internalized
- Workers therefore overvalue job security, relative to the social optimum
- As a result, a positive unemployment benefit financed by a lump sum tax on employed workers can improve welfare

Worker Skills

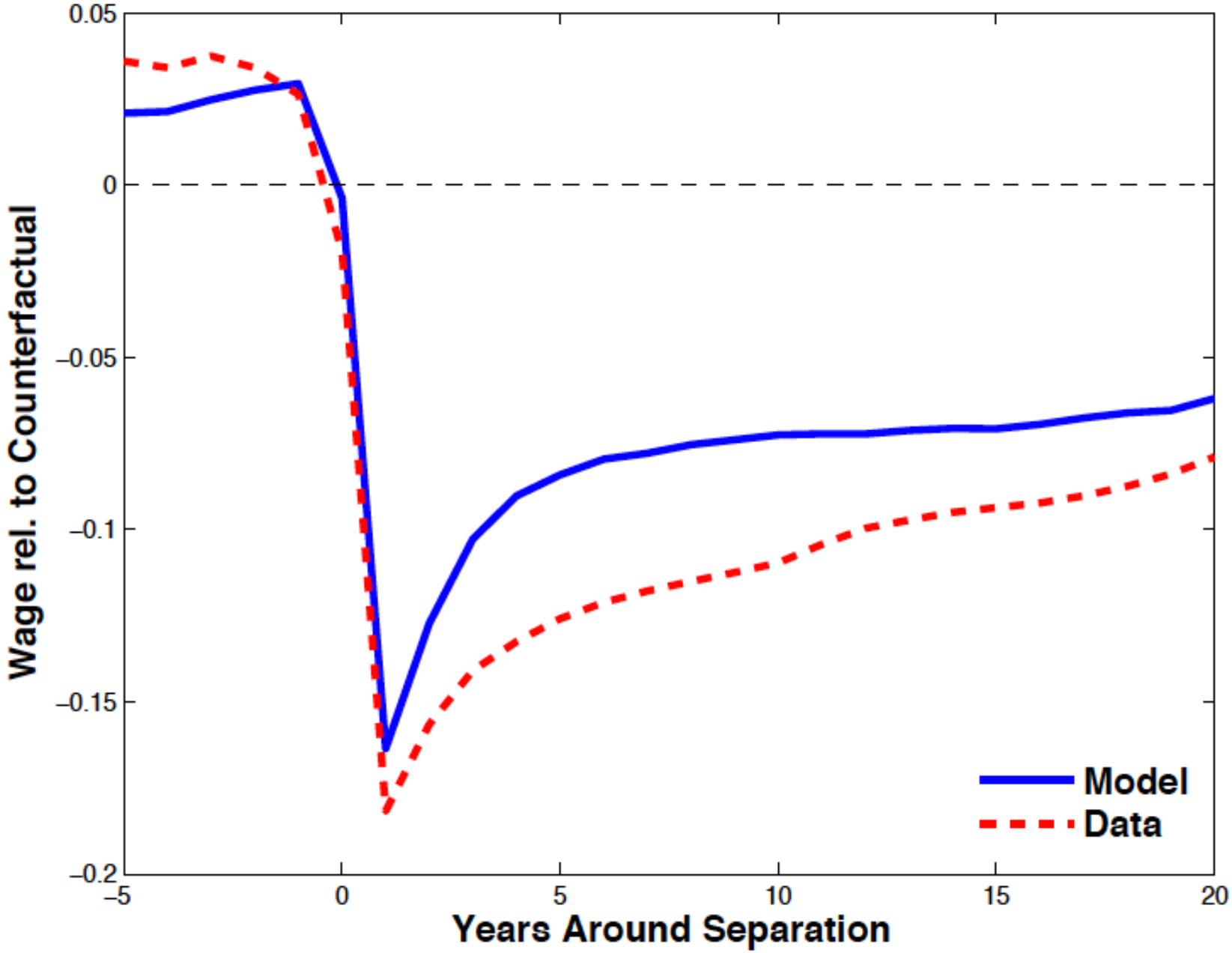
- Jarosch extends the model to add worker heterogeneity
- Workers have skill s, so productivity is $p(\theta_y, s)$
  - Increases probabilistically for employed workers
  - Depreciates probabilisitically for unemployed workers
- Still no permanent ex ante differences in workers

Estimation

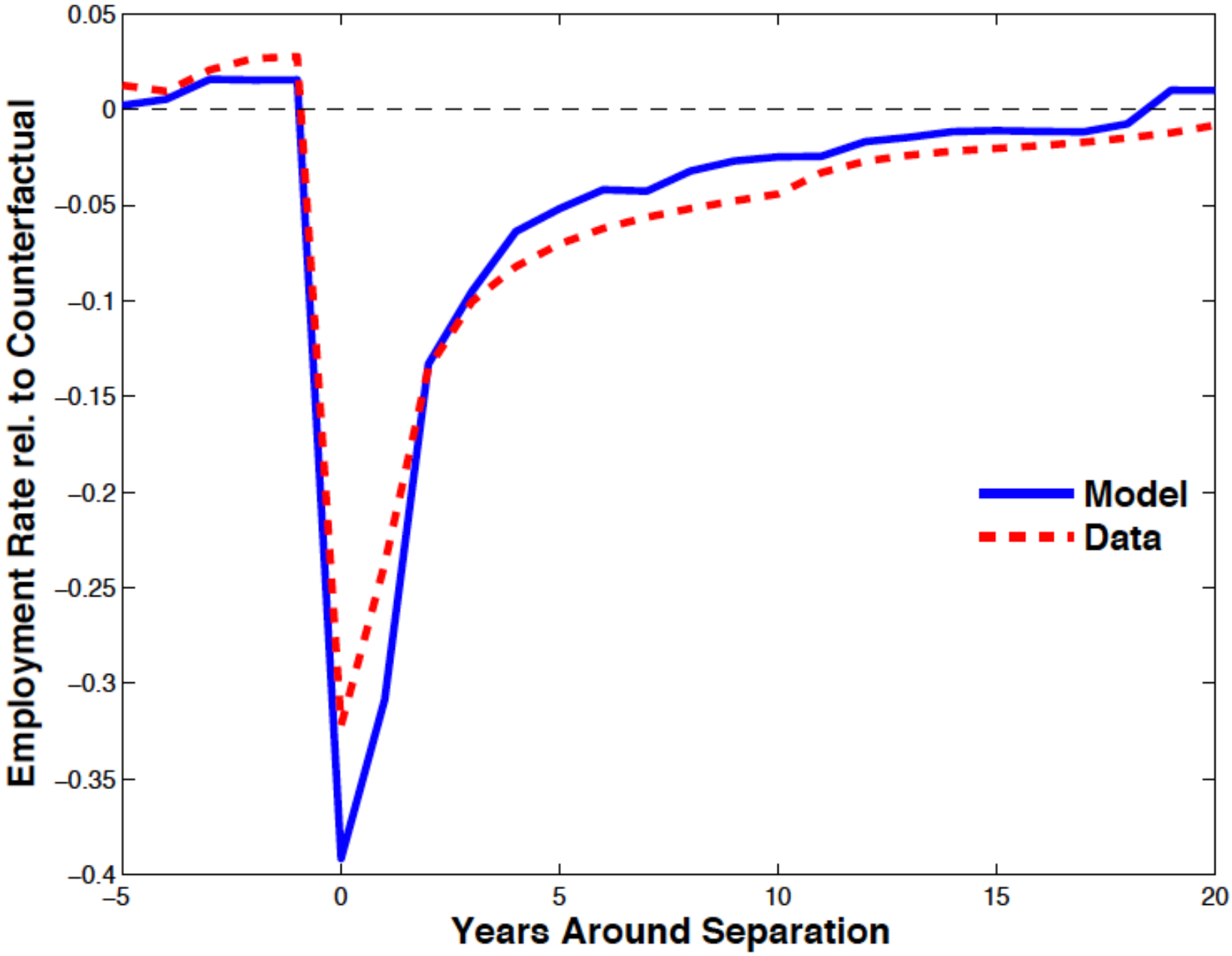
- Jarosch uses the model to quantify the importance of the job ladder, negotiation capital and experience effects in generating the displacement "scar"
- Matches moments from German social security data, 1974-2010
- Method of simulated moments estimator fitting various moments related to model parameters, e.g.:
  - Unemployment rate (λ_0)
  - Job-to-job transition rate (λ_1)
  - Unemployment hazard and slope w.r.t. tenure (distribution of θ_δ)
  - Relationship between wage and length of unemployment duration, w/worker FE (skill depreciation)
  - Relationship between wages and separation rate (correlation of θ_δ and θ_y)


0.05
0
−0.05
−0.1
−0.15
−0.2
−0.25
−0.3
−0.35
−0.4
−0.45
Earnings rel. to Counterfactual
−5
0
5
10
15
20
Years Around Separation
Model
Data

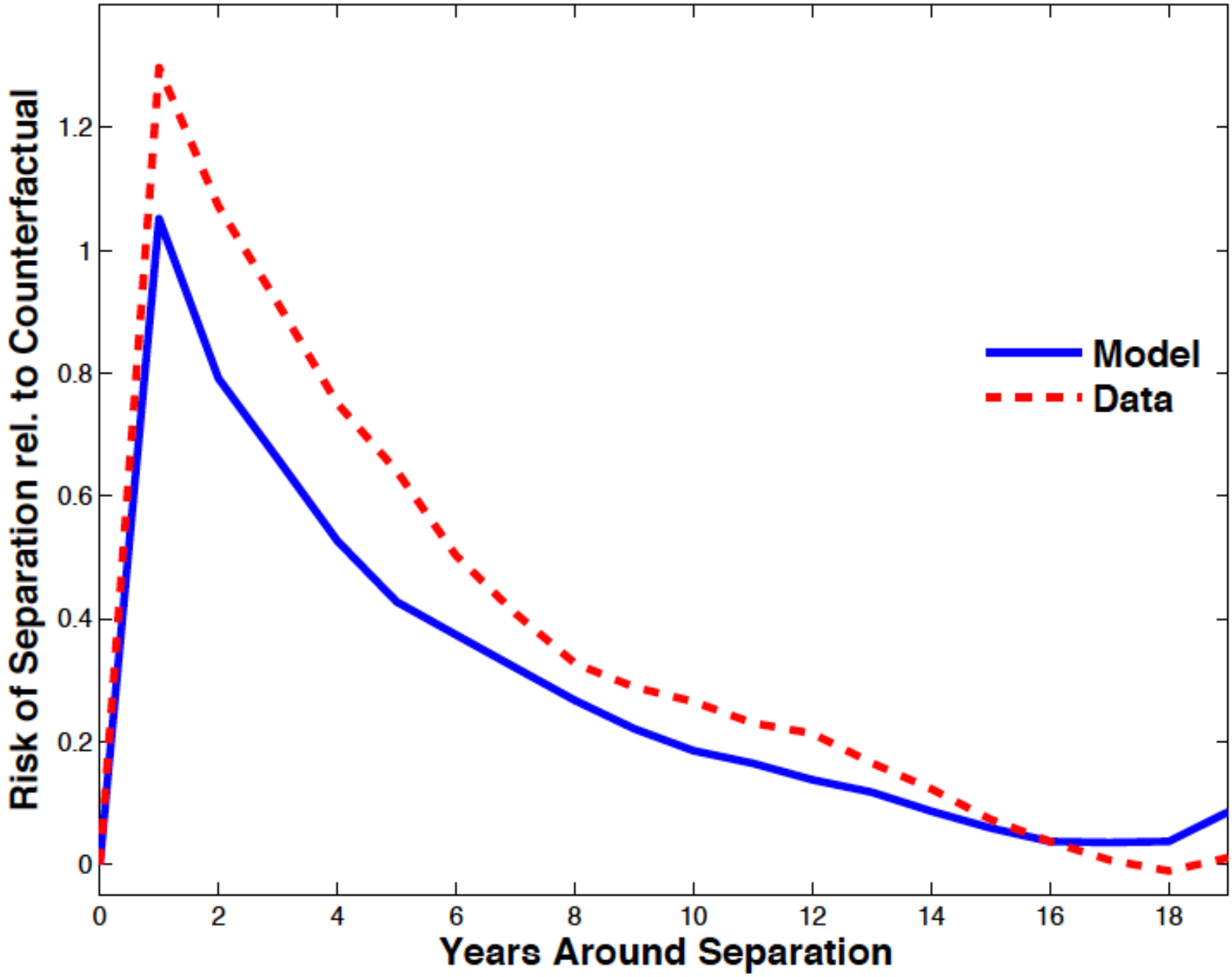
(a) Earnings


0.05
0
−0.05
−0.1
−0.15
−0.2
Wage rel. to Counterfactual
−5
0
5
10
15
20
Years Around Separation
Model
Data

(b) Wages



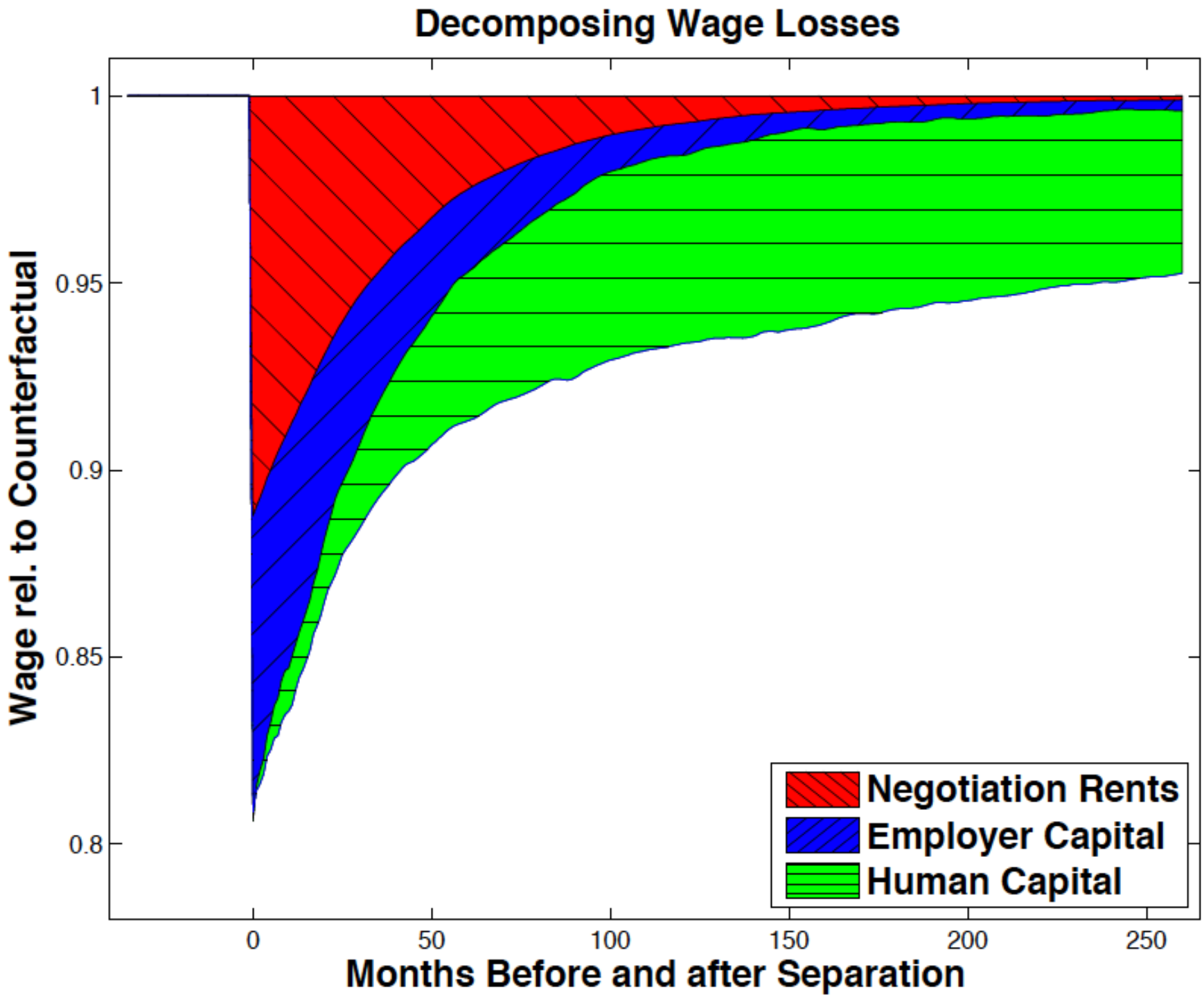
(c) Employment



(d) Job Security

Decomposition

- Three sources of wage scarring:
  - Negotiation rents
  - Position in job ladder (employer itself)
  - Human capital/experience
- Jarosch uses the model to "turn on" each component sequentially


Decomposing Wage Losses
Wage rel. to Counterfactual
1
0.95
0.9
0.85
0.8
0
50
100
150
200
250
Months Before and after Separation
Negotiation Rents
Employer Capital
Human Capital